

Causal Erasure: the Process-Rate Conditional Coordinate

GO-14 statement v0.7 — Theorems R, C and T (the convex program, and the n -transfer lemma), the (H*) *repair* — the cross-block read X , the lemma chain L1–L4, and the split-causal sub-family — T* v2, and **the plateau**: the Collapse identity, periodization, shift-averaging, and a two-sided certificate for the stationary value $\Psi(\Delta)$ that retires (H*), (H**), (H***), (D), (F) and the boundary charge from the plateau at $\Delta = 0, 1, 2$

Geometric Observation program

v0.7 — August 2026 (v0.6 superseded for the plateau: §6 proves the Collapse identity, step (A) periodization and step (B) shift-averaging, states Lemma S and Lemma C-stat, and certifies the stationary value two-sided by a *box-free per-frequency weak-duality* bound that needs no optimality of the fixed point, no convexity of the process-rate functional and no differentiability; the chain is *one-directional*, so what follows is $L^\infty(\Delta) \geq \Psi(\Delta)$ and not an equality, at margins $+0.0147817164/ + 0.0040574952/ + 0.0007968190$ over the *sealed* causal-spectral bars. Corollary 7 *supersedes* the v0.6 plateau arithmetic of Corollary 5 entirely; the (H**) and (D)+(F) numbers there are retained as *historical*. Also new: the certified $\Delta = 2$ finite- n anchors, the corrected block infimum, and the numbered errata R14–R20. v0.6 history: the H-repair lemma chain L1–L4 proves

$D_1 + D_2 \leq c(\Delta; m) + (2 - \mathbf{1}\{\Delta = 0\})X$ for *every* record of $\mathcal{F}_0$, X the cross-block read, using no optimizer structure at all; consequently (H*) is a *theorem* on the split-causal sub-family $\mathcal{F}_0^{\text{sc}}(m)$ with this document’s own constants, and is demoted to a *corollary* of the strictly weaker, better-margined, monotone-verified hypothesis (H**) on the single scalar X ; added: the Π -retraction lemma, the certified ϕ^{sc} brackets, the extended monotone $m \times \Delta$ table that retires the “ $m \in \{8, 16\}$ only” caveat, the repaired per-block feasibility step of T*(ii), the recomputed plateau corollary $0.5514005 > 0.5479448$, and the *named* obstruction that replaces the old repair-path remark. v0.5 history: Theorem T — exact subadditivity, Fekete existence of the limit and the *unconditional* lower transfer side — with (H*) carrying the boundary-charged reverse direction and the executed within-class certification of the diagonal ladder; six R-IND-5 transfer restatements in force, including the refutation of “ $\leq \kappa$ per side” as an $\mathcal{F}_0$ lemma. v0.4 history: Theorems R and C with the certified anchor table and the sandwich-margin interval, seven R-IND-5 restatements. v0.2 history: process-limit conjecture, hard hypothesis, dated erratum to Theorem 1(v); the v0.1 sign error was corrected at v0.2)

Abstract

GO-12 settled the single-record causal-prefix eraser (the Kalman family); the process-rate object — block records erased cell by cell, each against only the causal prefix of an aging reference — was promoted to a first theorem at v0.2 (netted, GO-P-2026-076). v0.3 added the process-limit face: the $n \rightarrow \infty$ limit of the coordinate exists at $O(1/n)$, closes at the smoothing-pole rate $\lambda_s^{2\Delta}$ with a constant that is *not* matched by any tested causal-conditioning spectrum,

and is driven by an explicit, testable record mechanism (horizon-matched V -cancellation). This revision adds the lower-bound theory: in moment coordinates (H, Γ) the family optimization is a jointly convex program — Theorem R (a schedule-general moment-coordinate representation of nL_a) and Theorem C (joint convexity, so every KKT point is the global family minimum and strong duality yields computable lower bounds). All seven full-space anchors are now *certified two-sided* (widths 3.45×10^{-6} – 1.7×10^{-5}), and the sandwich margin is a certified interval: $\min L_a(0) - \text{block}_{16} \in [0.0317872, 0.0317906]$. This revision adds the n -transfer theory. Theorem T is unconditional: block concatenation gives exact subadditivity $f(n_1 + n_2) \leq f(n_1) + f(n_2)$, so by Fekete the process limit $L^\infty = \lim_n \phi_n = \inf_n \phi_n$ exists, and the lower transfer side $0 \leq \phi_n - L^\infty$ is a theorem — every certified finite- n value bounds the limit from above. The *reverse* direction is not: the boundary charge is carried by a numbered hard hypothesis on dyadic-chain optimizers, because “ $\leq \kappa$ per side” is *false* as a family statement (D -feasible U -independent counterexamples reach 26κ ; no universal constant exists). The diagonal ladder is now certified *within class*. This revision *repairs* that hypothesis. Introducing the *cross-block read* $X = I(\hat{Y}^{b_1}; W^{b_2} \mid W^{b_1})$, a four-step lemma chain — κ is an *identity* not a bound, $I(E; T^{b_2}) \leq \kappa + X$, and the two per-side charge bounds — gives $D_1 + D_2 \leq c(\Delta; m) + (2 - \mathbf{1}\{\Delta = 0\})X$ for *every* record of $\mathcal{F}_0$, with no optimizer structure anywhere. Three things follow. (a) The refutation is *explained*: the counterexamples are exactly the records with enormous X (18.46 and 29.27 bits), and the bound $I(E; T^{b_2}) \leq \kappa + X$ is *tight* to 5.7×10^{-7} at one of them. (b) (H*) is a *theorem*, with this document’s own constants, on the split-causal sub-family $\mathcal{F}_0^{\text{sc}}(m)$ — a convex linear section, so Theorem C certifies it two-sided, at a split-causality value gap that is flat in n (≤ 0.0044 bits, and ≈ 0.0041 at $n = 16$), and onto which an explicit distortion- and block-law-preserving retraction Π maps every record. (c) The transfer results become conditional on (H**): *one scalar*, $X \leq \bar{X}$ at dyadic-chain optimizers, measured 6.3×10^{-3} against an admissible 0.117 (18 $\times$ headroom, against (H*)’s 5.5 $\times$) and monotone decreasing in m over $m \in \{8, 12, 16, 24\}$. The $\Delta = 0$ plateau survives the larger constant at $0.5514005 > 0.5479448$ (margin 3.456×10^{-3} , 203 $\times$ the quoted bracket width) and the base-24 negative control still fails, now by 3.11×10^{-4} . A separate hole in T*(ii), whose cost would have exceeded the whole repair term, is closed by convexity of the value function $D \mapsto \phi_m(D)$. Three further unconditional results are added. *Theorem K*: the KKT system of the convex program *splits by column block*, because $E[S \mid W] = V$ and Y are both coordinate blocks of W , so the Q -factor and the distortion gradient have disjoint supports; it gives $A_y = \theta N$, $N^{-1} = \theta I + V \Sigma_s^{-1} V^\top$, $A_v = -N V \Sigma_s^{-1} V_S^\top$, hence the first m -uniform regularity of the program ($0 \prec N \preceq I/\theta$, $0 \prec A_y \preceq I$, $\theta \geq 1.119$) and, as a *corollary*, the transpose symmetry that v0.3 could only measure. $\phi_{2m}^{\text{sc}} \leq \phi_m$ holds unconditionally, giving a third and preferable hypothesis (H***) on the split-causality value gap — preferable because that gap, unlike X , is *two-sided certifiable* by Theorem C (27 $\times$ headroom). And the repair path itself is now executed rather than promised: an m -uniform conversion theorem reduces (H**) to kernel-decay envelopes plus a noise floor, yielding $\bar{X} = 0.065038$ against an admissible 0.116930; what remains is named exactly — the operator whose inverse produces the optimizer is built *from* the optimizer, so inverse-closedness returns its own hypothesis, and a fixed-point argument in a weighted Banach algebra is the missing step. *This revision adds the plateau* (§6), and it retires most of the machinery above from that particular argument. An exact *Collapse identity* — $2 \ln 2 n L_a = \sum_t \ln \sigma_t - \ln \det N$ for every record, every n and every nondecreasing schedule — turns the two remaining steps into one-liners. *Step (A)*, periodization with *independent noise copies* (a load-bearing hypothesis: correlating them breaks the step outright), gives $L^{\text{process}}(D) \leq \phi_n$ for every n . *Step (B)*, shift-averaging, produces from any period- n record an *explicit* stationary record of no larger rate and identical per-symbol distortion. Both directions of both steps are inequalities the same way, so the chain is *one-directional*: it yields $L^\infty(\Delta) \geq \Psi(\Delta)$, the stationary spectral value, and it cannot yield the converse. Ψ is then certified *two-sided* by a box-free per-frequency weak-duality bound whose only inputs are admissibility of a pair of frozen filters, positivity of the per-frequency price,

and a correct cell infimum — no optimality of the fixed point, no convexity of the process-rate functional, no differentiability, and no moment box: $\Psi(0) \in [0.5627264963, 0.5627264964]$, $\Psi(1) \in [0.5364013784, 0.5364013785]$, $\Psi(2) \in [0.5310500198, 0.5310500199]$. Against the *sealed* causal-spectral allocation this is a plateau of $+0.0147817164/ + 0.0040574952/ + 0.0007968190$ at $\Delta = 0/1/2$ — four to eight orders above the certified bracket widths, at two lags the transfer route provably could not reach — *conditional on (R1) alone*, the convexity of the process-rate functional on the cyclostationary class, with (A)’s independent-noise-copy construction as the other named hypothesis. Two scoping results are load-bearing and stated as hard terms: (i) the record family — jointly Gaussian with (V, Y) , *independent of the reference noise U* — is a numbered hypothesis on the chain rule and on every minimum quoted here (U-coupled records break the identity and invert the sandwich); (ii) all numerical minima remain *class-conditional or finite-n upper bounds*; nothing here relies on them being global.

1 Setting

Sources as in GO-12: V AR(1) with pole a , unit stationary variance; $Y_t = \rho V_t + N_t$ with $\text{Var}(Y) = 1$; reference $S_t = V_t + U_t$, $\text{Var}(U) = \tau^2$. A block record $\hat{Y}^n$ of Y^n (per-symbol distortion $\frac{1}{n} \sum_t E(Y_t - \hat{Y}_t)^2 \leq D$) is erased cell by cell; the eraser of cell t holds $S^{t+\Delta}$ and the already-erased cells $\hat{Y}^{t-1}$. The coordinate is

$$L_a(\Delta) = \frac{1}{n} \sum_{t=1}^n I(T^n; \hat{Y}_t | S^{t+\Delta}, \hat{Y}^{t-1}), \quad T = (V, Y).$$

Hypothesis 1 (U-independence — hard, load-bearing). *The record family $\mathcal{F}_0$: $\hat{Y}^n$ jointly Gaussian with (V^n, Y^n) and independent of U^n given (V^n, Y^n) ; concretely $\hat{Y} = A_y Y + A_v V + Z$ with $Z \perp (V, N, U)$ (no U-coupling, $A_u = 0$). Theorem 1 (the chain rule and every item), Conjecture 1, and every minimum quoted in this document are stated under this hypothesis; no seal of the process-limit face is valid without it.*

Extension and counter-values (R-IND-5, on record). The chain rule of Theorem 1(i) extends *exactly* to U-coupled records $\hat{Y} = A_y Y + A_v V + A_u U + Z$ if and only if the U-coupling is Δ -lag causal, $A_u[t, s] = 0$ for $s > t + \Delta$. Outside that: the identity is *false* (measured residuals 0.015–0.40 at random U-couplings; 4.59 bits at an optimizer endpoint), and the minima collapse — the trivial Δ -causal U-record $\hat{Y}_t = 0.54 Y_t + 0.23 S_t + \mathcal{N}(0, 0.1625)$ is D -feasible and achieves $L_a(0) = 0.539536 < 0.566758$ (the family min), and a general U-coupled certified-feasible record achieves $L_a(0) = 0.092864 < \text{block}_{16} = 0.534968$: the sandwich of item (iii) is *inverted*. The U-coupled variant is a different, far cheaper coordinate and is *open*.

2 The theorem

Write $\lambda_s = a(1 - K)$ for the steady-state Kalman *smoothing* pole (K the steady Kalman gain of S on V); at the netted parameters $(a, \rho^2, \tau^2) = (0.8, 0.49, 0.4)$, $\lambda_s = 0.354554$, $\lambda_s^{-2} = 7.955$.

Theorem 1 (Causal erasure, first netting — GO-P-2026-076). *For every jointly Gaussian record on the window $t = 1, \dots, n$ satisfying Hypothesis 1:*

(i) **Exact chain rule.** $n L_a(\Delta) = I(T^n; \hat{Y}^n | S^n) + C_\Delta$ with

$$C_\Delta = \sum_t I(\hat{Y}_t; S_{>t+\Delta} | S^{t+\Delta}, \hat{Y}^{t-1}) \geq 0$$

the smoothing-leakage charge; $C_\Delta = 0$ iff the record is Δ -lag causally simulatable from S . The identity is exact for every record in $\mathcal{F}_0$ and every Δ (numerical residual $\sim 10^{-14}$); for the exact U -coupled extension and its failure outside the family see the counter-values above.

(ii) **Collapse is a feasibility face, not an impossibility.** Records that collapse ($C_\Delta = 0$) while carrying information exist — e.g. any V -free (N -only) record — so universal non-collapse is false as first conjectured. The correct statement: for $\tau^2 > 0$, $\Delta \leq n - 2$ and

$$D < \rho^2 \frac{n - \Delta - 1}{n},$$

every D -feasible collapsing record must be V -free on all cells $t \leq n - \Delta - 1$, which the distortion budget forbids; at $D \geq \rho^2(n - \Delta - 1)/n$ a feasible collapsing boundary record exists (exhibited at $\Delta = 6$, $n = 16$, $D = 0.3$). Strictness of $\min L_a > \text{block}/n$ in the sub-boundary regime routes through block-optimality plus $C_\Delta > 0$ at the (per-mode unique) block optimizer — not through universal non-collapse.

(iii) **Class-conditional sandwich, strictly decreasing.** With $\text{block}_n = \min \frac{1}{n} I(T^n; \hat{Y}^n | S^n)$ and $\text{slice}(\Delta)$ the netted single-letter slice coordinate,

$$\text{block}_n < \min_{\text{records}} L_a(\Delta) \leq L_a^{\text{diag}}(\Delta) < \text{slice}(\Delta),$$

where L_a^{diag} is the diagonal-class value; the upper bounds are strictly decreasing in Δ (v0.1 said increasing — a sign error: more prefix access can only cheapen causal erasure), and the lower margin closes geometrically at rate $\lambda_s^{2\Delta}$ (overstatement of the diag-class bound over the true minimum $\leq 1.07 \times 10^{-3}$ at $\Delta = 2$, $\leq 2.4 \times 10^{-6}$ at $\Delta = 5$; $n = 16$, $D = 0.3$).

(iv) **The smoothing pole — cell-local scope.** For time-local (cell-diagonal) records, the per-interior-cell leakage ratio $\text{leak}_t(\Delta)/\text{leak}_t(\Delta + 1) \rightarrow \lambda_s^{-2}$ in the window interior; the aggregate per-symbol ratio carries the finite-window cell-count prefactor $(n - \Delta - 1)/(n - \Delta - 2)$. The exponent is record-dependent: code-mixing records measurably decay slower (the reverse-water-filling record shows a stable $\approx 0.255/\text{lag}$, empirically $\approx \lambda_s^2/\rho^2$ — characterization open). No constant measured ratio is quoted — the pole is an asymptotic with a named prefactor and a named scope.

(v) **Not bookkeeping.** L_a is not the fixed-lag static quadratic: the deficit of the bookkeeping candidate is n -pinned, $\text{gap}_n = \text{static}(q_{\text{path}}) - \text{block}_n = 0.0213/0.0263/0.0280$ at $n = 8/16/24$, converging to the spectral gain ≈ 0.0313 ($\text{block}_\infty = 0.5299499808$ by per-frequency equal-slope allocation; twice-corrected value — see the dated erratum, Remark 10, and the R20 erratum of Remark 25). Record memory and cross-cell code structure carry real value.

Remark 1 (Class scoping — load-bearing). The diagonal (stationary-symmetric) class is provably suboptimal for L_a : at $\Delta = 0$, $n = 16$, $D = 0.3$ the first-order optimality certificate fails at

the diagonal optimum (feasible projected slope 8.6×10^{-3} against 10^{-6} noise) and explicit non-diagonal records achieve $\min L_a(0) \leq 0.567353 < 0.572255$ (an improvement of 4.9×10^{-3} bits, two independent parameterizations agreeing to 8×10^{-5}); the full-space search behind Conjecture 1 has since sharpened the beat to $\min L_a(0) \leq 0.5667581$ at $n = 16$ (Remark 8). Diagonal optimality *remains proven* for the block coordinate. Every quoted minimum is therefore a class-conditional or finite- n upper bound inside the bracket of (iii); presenting 0.572255 as *the* minimum would be an error.

Remark 2 (Rejected definition). Definition (b) (memoryless eraser) is monotone but has the wrong $\Delta \rightarrow \infty$ invariant (floors 0.0136 bits above the netted block coordinate) and is chain-rule-inconsistent; (c) (Kostina–Hassibi causally-conditioned DI) collapses to the full-path endpoint in Gaussian settings. Definition (a) is the thermodynamically operational coordinate pending a reset-protocol construction (open).

3 The convex program: Theorems R and C

Status: lower-bound prover + independent R-IND-5 verification (own evaluator, own polish, own moment box, fresh seeds; representation identity to 2.0×10^{-12} over 279 cells including edges; the composition chain of Theorem 2 re-proved independently; 11,200 verifier Jensen midpoints and 1,566 curvature probes with zero violations, on top of the prover’s 32,000). Harness: `experiments/go14.convexity.py`. Prereg 079 pending — nothing in this section is sealed yet.

Moment coordinates. Write $W = (V^n; Y^n)$ (so $T^n = W$), $\Sigma_W = \text{Cov}(W)$, and for a record $\hat{Y} = AW + Z$ of the family $\mathcal{F}_0$ set

$$H = \text{Cov}(\hat{Y}, W) = A\Sigma_W \quad (n \times 2n), \quad \Gamma = \text{Cov}(\hat{Y}) \quad (n \times n).$$

The family in these coordinates is the convex cone

$$\mathcal{D} = \{(H, \Gamma) : \Gamma - H\Sigma_W^{-1}H^\top \succeq 0\}$$

(the slack is $N_{\text{cov}} = \text{Cov}(Z)$), and the distortion is *affine*: $n \text{ dist} = \text{tr} \Sigma_Y - 2 \text{tr}(H \Sigma_W^{-1} \Sigma_{WY}) + \text{tr} \Gamma$. Write $P = \Sigma_W^{-1}$ and $Q = \Sigma_W^{-1} \Sigma_{WS} \Sigma_S^{-1} \Sigma_{WS}^\top \Sigma_W^{-1}$.

Theorem 1 (R — moment-coordinate representation, schedule-general). Assume Hypothesis 1. On the window $t = 1, \dots, n$ let the eraser of cell t hold $S^{se(t)}$ for an *arbitrary nondecreasing access schedule* $se(\cdot)$, and define the schedule-general counts

$$k(j) = \#\{t : se(t) < j\}, \quad 1 \leq j \leq n$$

(the number of record cells already erased when S_j enters the interleaved order). Then for every $(H, \Gamma) \in \text{int } \mathcal{D}$:

$$\begin{aligned} 2 \ln 2 \cdot n L_a &= \left[\ln \det(\Gamma - HQH^\top) - \ln \det(\Gamma - HPH^\top) \right] \\ &\quad + \sum_{j=1}^n \left[\ln \text{Var}(S_j \mid S^{j-1}) - \ln \text{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)}) \right]. \end{aligned}$$

Staircase reduction (explicit). For the coordinate’s own schedule $se(t) = \min(t + \Delta, n)$ the counts reduce to $k(j) = \max(0, j - \Delta - 1)$. The reduced form is *prefix-staircase-specific*: on non-staircase

nondecreasing schedules only the general $k(j) = \#\{t : se(t) < j\}$ is correct — blind use of the staircase formula deviates by up to 1.05 bits (verifier-measured), and off-by-one variants of k are detected at 0.1–0.7 bits.

Proof skeleton. (1) Factor the joint $\ln \det$ of $(S^n, \hat{Y}^n)$ along the interleaved order twice: collecting the $\hat{Y}$ -pivots gives $\sum_t \ln \text{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1})$; collecting the S -pivots gives $\sum_j \ln \text{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ — this regrouping is the whole trick (and the reason single-cell convexity is never needed). (2) U -independence enters *exactly once* as a premise — the residuals (U, Z) are independent given W — but is load-bearing in *two* legs of the computation: $\text{Cov}(S, \hat{Y}) = KH^\top$ with $K = \Sigma_{WS}^\top \Sigma_W^{-1}$ (numerator/leak leg, giving the Q -Schur complement $\Gamma - HQH^\top$ since $Q = K^\top \Sigma_S^{-1} K$), and $\ln \det \text{Cov}(\hat{Y} \mid W, S^\bullet) = \ln \det(\Gamma - HPH^\top)$ (denominator leg: conditioning on S adds nothing given W). (3) The unconditional S -pivots telescope against $\ln \det \Sigma_S$. This re-proves the 076 chain rule of Theorem 1(i) in one line: the bracket is the block term $I(T^n; \hat{Y}^n \mid S^n)$ and the j -sum is the leakage charge C_Δ in its S -side (reversed) form. Verification: residual 1.15×10^{-12} over 144 prover cells and 2.0×10^{-12} over 279 verifier cells including the edges $\Delta \in \{0, n-2, n-1, n\}$ and random non-staircase schedules. $\square$

Remark 3 (U -independence: one premise, two legs — and the non-conflation rule). Correcting only one leg does *not* restore the identity: at fixed dense- U -coupled records ($n = 12$) the leg-by-leg counter-values are 0.32–2.01 bits (as-stated, numerator-only- corrected, and denominator-only-corrected all break; the harness’s pinned record measures 2.68/3.51/0.83 bits respectively), while correcting *both* legs restores exactness for arbitrary A_u including anticausal couplings (residual at machine precision). Non-conflation: Δ -lag-causal U -coupling ($A_u[t, s] = 0$ for $s > t + \Delta$) extends the *record-space* chain rule of Theorem 1(i) exactly, but does *not* rescue the moment-form Theorem R as stated — measured moment-form residuals 2.4–3.5 bits at Δ -causal couplings (harness pinned record: chain-rule residual 2×10^{-16} , moment-form residual 3.22 bits at $\Delta = 2$). The two extension results must not be conflated.

Theorem 2 (C — joint convexity; KKT points are global; strong duality). On $\mathcal{D}$: (i) the block bracket $B(H, \Gamma) = \ln \det(\Gamma - HQH^\top) - \ln \det(\Gamma - HPH^\top)$ equals $-\ln \det(I - Z)$ with $Z = R^{1/2} H^\top (\Gamma - HQH^\top)^{-1} H R^{1/2}$ and

$$R = P - Q = \Sigma_W^{-1} \Sigma_{W|S} \Sigma_W^{-1} \succeq 0;$$

under the mapping $C \leftrightarrow H^\top$, $V \leftrightarrow \Gamma$ this is *exactly* the 074 convexity lemma’s G -form (GO-13 Theorem 5, block Schur lift), so Z is jointly matrix-convex on $\mathcal{D}$ with $Z \prec I$ on $\text{int } \mathcal{D}$ (Sylvester: $\det(I - Z) = \det M_P / \det M_Q$), and B is jointly convex. (ii) Each leak term $-\ln s_j$ is convex: the pivot $s_j = \text{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ is a Schur pivot of a joint covariance that is *affine* in (H, Γ) , hence an infimum of affine functions of (H, Γ) — concave — and bounded below by $\tau^2 > 0$ (the eraser never sees U_j through a U -independent record). (iii) The distortion is affine and $\mathcal{D}$ is a convex cone. Hence $n L_a$ is jointly convex on $\mathcal{D}$; every KKT point of the distortion-constrained program is the *global* family minimum; Slater’s condition holds (interior scalar records), so strong duality makes lower bounds computable: any $\mu \geq 0$ and any point x_p give the certificate $p^* \geq \phi_\mu(x_p) - \|\nabla \phi_\mu(x_p)\| R_{\text{box}}$, with R_{box} from the per-cell moment bounds $\Gamma_{tt} \leq (1 + \sqrt{nD})^2$, $|H_{tj}| \leq \sqrt{\Gamma_{tt}}$ (verifier: the box is valid and nearly tight — an adversarial one-cell record reaches 10.1748 against the box’s 10.1818).

Proof skeleton. (i) is the 074 lift verbatim after the transpose-convention check ($C^\top P C$ with $C = H^\top$ is $HPH^\top$ — mapping exact to 0.0); $R \succeq 0$ by the displayed closed form (verified to 1.7×10^{-14} ; $\text{eigmin}(Q) \geq -10^{-15}$, PSD to machine). (ii) is inf-of-affine concavity plus $-\ln$ composition on

$[\tau^2, \infty)$. (iii) is inspection. Numerics on record: zero Jensen-midpoint violations (prover 32,000 + verifier 11,200 + 1,566 curvature probes), tangent plane at the (16,0) winner 400/400 with min slack $+1.13 \times 10^{-2}$, analytic gradient vs FD 1.2×10^{-10} . $\square$

Remark 4 (Per-cell convexity is open — the regrouping is the proof). Theorem 2 proves convexity of the *regrouped* pieces (block bracket, S -side leak sum). Whether the individual per-cell CMI terms of L_a are themselves convex in (H, Γ) remains *open*: no violation was found in any sampling (prover and verifier both, zero hits), but no proof exists either. Nothing in this section relies on per-cell convexity.

Remark 5 (The diagonal class is a convex linear section — its ladder is UB-only because class $<$ family). In (H, Γ) the diagonal (stationary-symmetric) class is a *convex (linear) section* of $\mathcal{D}$: midpoints of diagonal records stay in class (verifier: off-diagonal mass in the C_v eigenbasis at (H, Γ) -midpoints is 0 to machine precision). The earlier “non-convex slice” wording is refuted as worded — what is non-convex is the (g, h, z) *parameter chart* (explicit midpoint witness on record), a different statement. Consequently the diagonal ladder of §4 is upper-bound-only *because the class is a strict subset of the family* (Remark on class scoping, §2), not because of any slice non-convexity; and within-class values are certifiable by the same Lagrangian machinery restricted to the section. **That upgrade path has since been executed**: all nine ladder cells and three block values are now certified two-sided *class* values (§5.2). Certifying the class does not certify the family — the ladder remains an upper bound on $\min L_a$ because class $<$ family.

Certificates on record (per- n). $\text{eigmin}(P - Q) = 1.0382 \times 10^{-2} / 5.5944 \times 10^{-3} / 4.4309 \times 10^{-3} / 3.9529 \times 10^{-3}$ at $n = 8/16/24/32$ — *decreasing in n* ; quoted as “ $\geq 3.9 \times 10^{-3}$ for $n \leq 32$ ” (no n -uniform floor is claimed; the earlier single-number quote is an erratum, corrected here). The noise-floor parameterization is inactive at every winner and every polished point.

3.1 Theorem K: the KKT system splits by column block

Two *exact* structural facts of this model drive everything in this subsection, and both are checkable in one line:

$$K = \Sigma_{WS}^\top \Sigma_W^{-1} = [I \ 0], \quad \Sigma_W^{-1} \Sigma_{WY} = [0; I]$$

(verified to 2×10^{-15}): the first because $E[S \mid W] = V$ and V is a coordinate block of W , so every Q -factor term lives only on the V -columns of $A = H \Sigma_W^{-1} = [A_v, A_y]$; the second because Y is a coordinate block of W , so the distortion gradient lives only on the Y -columns. **The two supports are disjoint**, so the stationarity condition splits by column block.

Theorem 3 (K — KKT structure; *unconditional*, $\mathcal{F}_0$, any Δ , any nondecreasing schedule). Let (H, Γ) be a KKT point of the distortion-constrained program with multiplier μ , and write $N = \Gamma - H \Sigma_W^{-1} H^\top = \text{Cov}(Z)$, $\theta = 2\mu \ln 2$, and (v_t, v_t^S, σ_t) for the *record* pivots of the interleaved order (cell t predicted from $S^{\text{se}(t)}, \hat{Y}^{t-1}$), collected into $V \Sigma_s^{-1} V^\top$ and $V \Sigma_s^{-1} V_S^\top$. Then

$$(K1) \ A_y = \theta N, \quad (K2) \ N^{-1} = \theta I + V \Sigma_s^{-1} V^\top, \quad (K3) \ A_v = -N V \Sigma_s^{-1} V_S^\top.$$

Verified at every certified optimizer — relative residuals $\leq 3.2 \times 10^{-8}$ at (16,0), (24,0), (32,0), (48,0), (16,1), (16,2), (24,1) (prover), and $\leq 8.3 \times 10^{-8}$ over the six of those the harness recomputes from cold starts. The residuals track the polish accuracy of the optimizer, not the identities, which are exact; the harness gates them at 10^{-6} for that reason.

Corollaries, unconditional and m -uniform.

- (a) $A_y = \theta N$ is *symmetric positive definite*. **This upgrades the 078-era “mechanism” finding — A_y and the noise kernel transpose-symmetric, all causal asymmetry confined to A_v — from an empirical observation to a corollary of Theorem 3** (measured symmetry defect 1.2×10^{-9} – 1.4×10^{-8} , against the 5×10^{-5} at which it was first reported; see Conjecture 1(4)). And (K3) carries the causal asymmetry through V_S , whose column support ends *exactly* at $t + \Delta$ — the structural candidate for the observed sign change at lag Δ . *This is a lead, not a claim*: no proof that the sign change must occur there is offered here.
- (b) $0 \prec N \preceq I/\theta$ and $0 \prec A_y \preceq I$, *uniformly in m* — the first m -uniform optimizer-regularity statement in the program.
- (c) $\theta \geq 2 \ln 2 \phi_n(D)/(1 - D) \geq 1.119$ (convexity of $D \mapsto \phi_n(D)$, Remark 14, together with $\phi_n(1) = 0$), hence $N \preceq 0.894 I$ unconditionally.

Proof skeleton. Stationarity in Γ reads $M_Q^{-1} - N^{-1} + \partial_\Gamma(-\sum_j \ln s_j) = -\theta I$. The Cholesky decomposition of J^{-1} along the interleaved order, $J = \text{Cov}(S^n, \hat{Y}^n)$, splits the pivot Gram into a reference half and a record half summing to J^{-1} (verified to 6×10^{-16}); since $(J^{-1})_{\hat{Y}\hat{Y}} = M_Q^{-1}$, the M_Q^{-1} and reference-pivot terms combine into the record-pivot Gram — which is (K2). Stationarity in H then splits by the disjoint supports above: the Y -columns give $N^{-1}A_y = \theta I$, i.e. (K1); the V -columns give (K3). *Caution on reading (K2) and (K3)*: the Gram must be the *record*-pivot one — substituting the reference-pivot Gram makes (K2) false at the 47% level and (K3) false at the 120% level. $\square$

3.2 Certified two-sided anchors

Lagrangian polish + moment-box tangent bound (Theorem 2), prover (P) and independent verifier (V) each end-to-end; per anchor the tighter certified width of the two is cited. Upper ends are the (feasibility-projected) winner values.

(n, Δ)	certified bracket for p^*	width	source
(16, 0)	[0.566754682, 0.5667581350]	3.45×10^{-6}	V
(16, 1)	[0.5408029, 0.5408064658]	3.6×10^{-6}	P
(16, 2)	[0.5358901, 0.5358939352]	3.8×10^{-6}	P
(24, 0)	[0.5654096, 0.5654138570]	4.3×10^{-6}	P
(24, 1)	[0.5393228, 0.5393377781]	1.5×10^{-5}	P
(24, 2)	[0.5342628, 0.5342788038]	1.6×10^{-5}	P
(32, 0)	[0.5647249, 0.5647418676]	1.7×10^{-5}	P

The 078 anchors are thereby *certified family values*, no longer upper bounds alone; minimizer *uniqueness* is still not claimed (no flat was found — three polishes agree to 2.8×10^{-14} in value and 2×10^{-6} in coordinates — but gauge flats are not excluded; “an optimizer”, never “the”). The harness reproduces four of the brackets end-to-end from cold starts by μ -bisection (convexity makes warm starts a convenience, not a dependence), at widths $4.0/3.0/3.8/9.2 \times 10^{-6}$ for (16, 0)/(16, 1)/(24, 0)/(32, 0) and a block_{16} bracket of width 1.7×10^{-6} .

Corollary 1 (Certified sandwich margin at $n = 16$, $D = 0.3$).

$$\min L_a(0) - \text{block}_{16} \in [0.0317872, 0.0317906]$$

(verifier-tightened interval). The block side inherits the sealed block_{16} precision through the proven diagonal optimality of the block coordinate.

Remark 6 (Feasible projection of winner upper ends). The stored winner endpoints are distortion-infeasible by $\sim 4 \times 10^{-11}$ (raw optimizer termination). The one-line feasible projection (blend toward the near-copy record) corrects the upper ends by $\sim 10^{-10}$ — immaterial at the 3.5×10^{-6} width scale; all upper ends above are to be read through this remark.

Remark 7 (“Certified” means floating-point certificates). All brackets here are *floating-point* certificates: the weak-duality and tangent-bound inequalities are exact in real arithmetic and are evaluated in IEEE double precision without interval arithmetic. The certified widths exceed the $f64$ evaluation error by roughly seven orders of magnitude; no claim of formally verified arithmetic is made.

4 The process limit

Everything in this section is under Hypothesis 1 and at the netted parameters, $D = 0.3$. Status: probe + independent R-IND-5 verification (two evaluator routes agreeing to 2×10^{-10} ; all sealed 076 anchors reproduced); pre-registration (078) pending — no numerical face here is sealed yet.

Conjecture 1 (Process limit of the causal-erasure coordinate — v0.2, R-IND-5-scoped). *Let $L_n^{\text{fs}}(\Delta) = \min L_a(\Delta)$ over the full (non-diagonal) family $\mathcal{F}_0$ on the window of length n , and $L_n^{\text{diag}}(\Delta)$ the diagonal-class value.*

- (1) **Existence at $O(1/n)$.** Both coordinates converge as $n \rightarrow \infty$ with $O(1/n)$ finite-window drift (model attack: fitted exponent $\alpha = 0.996\text{--}1.001$). Two-point $1/n$ limits, calibrated on the block face ($\leq 5 \times 10^{-6}$): full space $L_\infty^{\text{fs}}(\Delta) = 0.562725/0.536400/0.531049$ at $\Delta = 0/1/2$ (three- n consistency at $\Delta = 0$: $0.5627253/0.5627259/0.5627265$ from $n = 16/24/32$ pairs); diagonal class $0.568571/0.537401/0.531206/0.529977$ at $\Delta = 0/1/2/4$ ($\pm \leq 1.2 \times 10^{-5}$).
- (2) **Closure rate, constant unclaimed.** The excess $E_\infty(\Delta) = L_\infty^{\text{diag}}(\Delta) - \text{block}_\infty$ obeys $E_\infty(\Delta) \sim c \lambda_s^{2\Delta}$ as a $\Delta \rightarrow \infty$ asymptotic only: the per-lag ratio rises toward $\lambda_s^{-2} = 7.955$ from below ($5.18 \rightarrow 5.93 \rightarrow 6.87 \rightarrow 7.65 \rightarrow$ within error at $\Delta = 5 \rightarrow 6$) and is unresolved beyond $\Delta \approx 6$; the constant is quoted as the bracket $c_{\text{diag}} \in [0.105, 0.125]$ only — $c(\Delta) = 0.107/0.112/0.117$ at $\Delta = 4/5/6$, still rising, not converged through $\Delta = 6$. The full-family constant is data-supported only as $c_{\text{fs}} \in (0.07, 0.125]$. No single-letter identification of c is claimed.
- (3) **Strictly above every tested causal-conditioning spectrum — certified at every computed n ; in the limit conditional on (H^*) at $\Delta = 0$ and on the $O(1/n)$ model at $\Delta = 1, 2$.** Both limit conditions are discharged at v0.7 — **Corollary 7** derives the limit statement at $\Delta = 0, 1, 2$ from §6, conditional on (R1) alone; what follows in this item is the v0.4-era route and is retained as history. At every computed anchor ($n \in \{16, 24, 32\}$, $\Delta \in \{0, 1, 2\}$) the certified lower bracket end of §3.2 sits strictly above

the causal-spectral allocation value (Remark 9) — a certificate-strict gap (smallest exceedance $\approx 4 \times 10^{-3}$ bits at $(24, 2)$, $\geq 250 \times$ the bracket width there), not an optimizer report. In the limit, the plateaus $+0.014781/ + 0.004057/ + 0.000795$ ($\Delta = 0/1/2$) above the allocation are $160\text{--}3000 \times$ the calibrated extrapolation error and carry the wrong shape (drifting per-lag rate, not the allocation's clean λ -law). At $\Delta = 1, 2$ the limit statement remains conditional on the $O(1/n)$ finite-window model. At $\Delta = 0$ it is no longer model-conditional: Corollary 5 derives $L^\infty(0) \geq 0.5514005 > 0.5479448$ from the certified $n = 32$ bracket and the repaired transfer constant, conditional instead on Hypothesis 2 (H^{**}) — or on $(D)+(F)$, at the smaller margin recorded in Corollary 5. The subadditivity half of the old n -monotonicity face is now closed unconditionally (Theorem 4); what remains open is the cross-block read at optimizers (Question 1). Both closed-form candidates ($c_{\text{spec}} = 0.0190 \pm 0.0007$, $c_{\text{static}} = 0.026668$) are too small at every Δ . No claim is made that no single-letter form exists. The gap is genuine cross-cell code value: the winner cuts leakage $0.033 \rightarrow 0.024$ by paying block penalty the per-frequency relaxations cannot express.

- (4) **Mechanism — the symmetry half is no longer a conjecture.** At every full-space optimizer computed ($n = 16/24/32$, $\Delta = 0/1/2$; seven adversarial starts converge to the same record, KKT certificate 8.4×10^{-6}): the Y -coupling and noise kernels are time-symmetric, where time-symmetric means transpose symmetry $A_y = A_y^\top$ (relative residual $< 5 \times 10^{-5}$ as first measured; the stronger persymmetry $JA_y^\top J = A_y$ fails at the $\approx 6\%$ level at the window edges — transpose symmetry is the invariant); all causal structure lives in the V -coupling A_v , whose interior-row kernel changes sign exactly at lag Δ — horizon-matched V -cancellation: the record aligns V -content with the span the eraser will hold and anti-correlates V beyond the access horizon, suppressing smoothing leakage. This is exactly what the symmetric-by-class diagonal family cannot express (and explains the 076 beat). **Status upgrade (v0.6).** The first half of this item is no longer conjectural: $A_y = \theta N$ is symmetric positive definite at every KKT point by Theorem 3(K1), so the transpose symmetry of A_y and of the noise kernel, and the confinement of all causal asymmetry to A_v , are corollaries, not measurements (measured symmetry defect now 1.2×10^{-9} – 1.4×10^{-8} , i.e. at polish accuracy). What remains conjectural is the sign change exactly at lag Δ ; Theorem 3(K3) supplies a structural candidate — A_v inherits its asymmetry from V_S , whose column support ends exactly at $t + \Delta$ — but no proof. Falsifiable per-cell at any n .

Remark 8 (Reproducible anchors — raw finite- n values only). The anchors of record are raw finite- n L_a values: family $\min L_a(0) \leq 0.5667581$ at $n = 16$ (beats the 076-recorded 0.567353 by 5.9×10^{-4} ; all start basins converge to it within 10^{-7}); full-space $L_a(0) \leq 0.56474187$ at $n = 32$; the diagonal ladder value $L_a^{\text{diag}}(n=48, \Delta=6) = 0.5316222730$. The full-space upper ends are now *certified two-sided* (§3.2); the diagonal-ladder values are now certified two-sided *within the class* (§5.2) and remain class-conditional upper bounds on the family minimum (Remark 5). The extrapolated $E_\infty(6)$ is quoted *only* with an explicit error bar — $E_\infty(6) \approx 4.4 \times 10^{-7} \pm 10\%$ (two-point pair sensitivity); it is *not* a four-digit anchor, the raw $L_a(48, 6)$ above is.

Remark 9 (The causal-spectral allocation is a reference construction). The causal-spectral allocation: equal-slope per-frequency conditional quadratics with conditioning spectrum the Δ -lag causal Wiener error spectrum $S_e^{(\Delta)}$. Values $0.547945/0.532344/0.530253$ at $\Delta = 0/1/2$, verified to seven digits by an independent (time-domain) route; it reproduces block_∞ at the two-sided endpoint to 1.6×10^{-10} , and its own excess obeys the $\lambda_s^{2\Delta}$ law cleanly. It is a *reference construction only*: it

carries no achievability map (not an upper bound on $\min L_a$) and no data-processing direction (not a lower bound). Item (3) of Conjecture 1 is a statement about mismatch with the tested spectra, never a bound.

Remark 10 (Erratum — dated 2026-08-05). v0.2 of this document quoted $\text{block}_\infty = 0.52991$ in Theorem 1(v), propagated from the pass-1 coarse frequency waterfilling; the converged per-frequency equal-slope value is $\text{block}_\infty = 0.529950$ (corrected in item (v) above; the spectral gain ≈ 0.0313 is unaffected). That six-digit value is itself superseded at the ninth digit by the anchor-free two-sided certificate of Remark 25(R20): $\text{block}_\infty = 0.5299499808119$. Same date, probe-record erratum: the process-limit probe's diagonal excess at $\Delta = 4$ is 2.66×10^{-5} , not 2.75×10^{-5} .

Dated 2026-08-06 (transfer note). Three further corrections, all carried into §5: (a) the constant was recorded as $c(1) \leq 1.1257$; the computed value is 1.1257294, so the recorded bound is false in the fifth decimal and $c(1) \leq 1.1258$ is used throughout (Remark 15). (b) The transfer note cites a value 0.564729705 for the certified lower end at $(32, 0)$; no computation on record produces it. The sealed value is $\text{LB}(32, 0) = 0.5647327869$ (`results/G014-convexity.json`, `s5_32_0`), and it is the only one used in Corollary 5; 0.564729705 must not be cited. (c) The plateau margin is $215\times$ the quoted bracket width, not the $\approx 250\times$ of the transfer note.

5 The n -transfer lemma: Theorem T, and the repair of (H^*)

Status: prover + independent R-IND-5 verification (own `slogdet` evaluator, own certified anchors, own re-derivation of the constants and of the corollary arithmetic; six mandatory restatements, all in force in this revision). Harness `experiments/go14_transfer.py`. Registration 080 pending — nothing in this section is sealed. The repair material of §5.1 (the lemma chain, the split-causal sub-family, the Π -retraction, the monotone table, the feasibility repair and the recomputed corollary) and Theorem 3 of §3.1 are newer still: they carry their own harness, `experiments/go14_hrepair.py`, and await registration 081. **Nothing in this revision contradicts a gate of 080:** every 080 gate is either untouched or — in the case of the κ -per-side refutation — *explained* rather than overturned.

Write $\phi_n(\Delta) = \inf_{\mathcal{F}_0} L_a(\Delta)$ for the family value on the window of length n (certified two-sided in §3.2) and $f(n) = n \phi_n(\Delta)$. Let

$$\kappa = \frac{1}{2} \log_2 \frac{1}{1-a^2} = 0.736966 \text{ bits}$$

be the AR(1) boundary information at the netted pole $a = 0.8$. For a split of the n -window into blocks b_1, b_2 define the *per-side boundary charges*

$$D_i = \sum_{t \in b_i} \left[I(T^{b_i}; \hat{Y}_t \mid \text{own}C_t) - I(T^{b_i}; \hat{Y}_t \mid \text{big}C_t) \right] \geq 0,$$

where $\text{own}C_t$ is the conditioning set of cell t inside its own block and $\text{big}C_t = S^{se(t)}, \hat{Y}^{t-1}$ is the big-window one; D_i is exactly what the split loses at the boundary. Write $E = (S^{b_1}, \hat{Y}^{b_1})$ for the block-1 record-and-reference history that the block-2 cells see in $\text{big}C$ but not in $\text{own}C$.

Theorem 4 (T — exact subadditivity, Fekete existence, and the lower transfer side; *unconditional*). Fix Δ and assume Hypothesis 1.

- (i) **Exact subadditivity.** $f(n_1 + n_2) \leq f(n_1) + f(n_2)$ for all $n_1, n_2 \geq 1$. Take ε -optimal records of the two windows, independent of each other, and concatenate: the result is a feasible record of the $(n_1 + n_2)$ -window (the distortion constraint is the cell average), and *at every cell* the big-window per-cell term is at most the own-window term. The numerator $\text{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1})$ can only shrink when the conditioning set grows from the own block to the whole window; the denominator $\text{Var}(\hat{Y}_t \mid T^n, S^{se(t)}, \hat{Y}^{t-1})$ is *unchanged*, because a block-diagonal record makes cell t conditionally independent of the other block given its own block's T — verified as an exact equality (relative deviation 2×10^{-14} over all cells, including the unequal split $8 + 4 \rightarrow 12$). Letting $\varepsilon \downarrow 0$ gives the inequality for the infima.
- (ii) **Fekete.** $\phi_n \geq 0$, because each per-cell term is a conditional mutual information; hence $f \geq 0$ is subadditive and

$$L^\infty(\Delta) := \lim_{n \rightarrow \infty} \phi_n(\Delta) = \inf_n \phi_n(\Delta)$$

exists in $[0, \infty)$.

Hygiene (both steps were demanded by R-IND-5 and are stated, not assumed). (a) The non-negativity $\phi_n \geq 0$ is what makes Fekete's lemma give a finite limit rather than $-\infty$; it is immediate from the CMI form of the coordinate, but it is a hypothesis of the lemma and is recorded as such. (b) The concatenation argument is run on ε -optimal records and the limit $\varepsilon \downarrow 0$ is taken at the end, so the theorem never needs ϕ_n to be *attained*. (Attainment does hold, by Theorems 1 and 2 plus the moment box, but it is not used.)

Corollary 2 (The lower transfer side is a theorem). $0 \leq \phi_n(\Delta) - L^\infty(\Delta)$ for every n — *unconditionally*. Every certified finite- n family value of §3.2 is therefore an upper bound on the limit; in particular $L^\infty(0) \leq \phi_{32}(0) \leq 0.5647419677$.

5.1 The reverse direction: the cross-block read X

The matching *upper* transfer bound $\phi_n - L^\infty \leq c/n$ needs the boundary charge to be controlled at the optimizers of the doubling chain. At v0.5 that control was a bare hypothesis, because “ $\leq \kappa$ per side” is refuted as a family lemma (Remark 12). This revision *repairs* it: a lemma chain, using no optimizer structure whatsoever, reduces the whole reverse direction to a single scalar.

Definition 1 (The cross-block read). For a split of the n -window into $b_1 = \{1, \dots, m\}$ and $b_2 = \{m+1, \dots, n\}$ write $W^{b_i} = T^{b_i} = (V^{b_i}, Y^{b_i})$, $E = (S^{b_1}, \hat{Y}^{b_1})$ and $E' = S^{b_2, 1.. \Delta}$ (empty at $\Delta = 0$). The *cross-block read* of a record is

$$X := I(\hat{Y}^{b_1}; W^{b_2} \mid W^{b_1}) \geq 0,$$

the information the block-1 *rows* draw from block-2 sources beyond what block-1 sources already say. $X = 0$ exactly when the record is *split-causal* at m (Definition 2).

Lemma 1 (L1 — κ is an *identity*, not a bound). $I(T^{b_1}; T^{b_2}) = \kappa$ exactly, *at every split and every n* . Proof. V is $AR(1)$, hence *Markov*, so $T^{b_1} \rightarrow (V_m, V_{m+1}) \rightarrow T^{b_2}$ and both data-processing steps are equalities; the surviving term is $I(V_m; V_{m+1}) = \frac{1}{2} \log_2 \frac{1}{1-a^2} = \kappa$. Certificate: $\max_m |I - \kappa| = 3.8 \times 10^{-14}$ over $m \in \{3, 4, 6, 8, 12, 16\}$, and the spread over m is 4.1×10^{-14} — the proof predicts a constant, and no drift in m is observed. $\square$

Lemma 2 (L2 — the cross-read bound, and it is sharp). $I(E; T^{b_2}) \leq \kappa + X$ and $I(E'; E) \leq \kappa + X$. Proof. S^{b_1} drops out of the numerator because $U \perp (W, \hat{Y})$: conditionally on W^{b_1} the reference noise carries nothing about T^{b_2} . What is left is $I(W^{b_1}; T^{b_2}) + I(\hat{Y}^{b_1}; W^{b_2} \mid W^{b_1}) = \kappa + X$ by Lemma 1 and Definition 1; the second leg is the same computation with E' in place of T^{b_2} . $\square$ Sharpness (this is the point). Over the pinned battery — generic $\mathcal{F}_0$ records, split-causal records, and the two adversarial copy-row classes of Remark 12 — the minimum slack is $+5.67 \times 10^{-7}$, attained at the two-V-copy counterexample, i.e. the bound is tight exactly where the universal claim dies. (The second leg's minimum slack is $+0.365$.)

Lemma 3 (L3 — the block-2 charge). $D_2 \leq I(E; T^{b_2}) - I_m$. Proof. The interleaved chain rule gives the exact identity

$$D_2 = I(E; T^{b_2}) - I(E; T^{b_2} \mid S', \hat{Y}') - \sum_j I(E; S'_j \mid \text{pfx}_j),$$

whose first subtracted term is a mutual information (≥ 0); the second sum has every term ≥ 0 , so it is at least its partial sum over $j \leq \Delta + 1$, which telescopes (there $\text{pfx}_j = S'^{1..j-1}$, no record cell having been erased yet) to $I(E; S'^{1..\Delta+1}) \geq I_m$ by data processing ($E \supseteq S^{b_1}$). Certificates: identity residual $\leq 9.3 \times 10^{-11}$, chain-rule collapse 6.7×10^{-16} , $\min(I(E; S'^{1..\Delta+1}) - I_m) = +1.47 \times 10^{-2}$ over the battery, and the resulting bound holds over the whole battery with minimum slack $+0.528$. The zero-claim of Remark 16 enters here and is $\leq 2.5 \times 10^{-10}$ on $\mathcal{F}_0$. $\square$

Lemma 4 (L4 — the block-1 charge). $D_1 \leq I(E'; E)$, and $D_1 = 0$ exactly at $\Delta = 0$ (the block-1 extension set is empty — measured 0 to 0.0). $\square$

Theorem 5 (H — the repair inequality; unconditional on $\mathcal{F}_0$). For every record of $\mathcal{F}_0$ and every split at m ,

$$D_1 + D_2 \leq c(\Delta; m) + (2 - \mathbf{1}\{\Delta = 0\})X \quad c(\Delta; m) = (2 - \mathbf{1}\{\Delta = 0\})\kappa - I_m,$$

$I_m = I(S^{b_1}; S'^{1..\Delta+1})$: add Lemmas 3 and 4, and bound each right-hand side by Lemma 2. No optimizer structure is used anywhere in the chain — it is a deterministic, analytic statement about the family. Over the pinned battery (22 cells, including both refuting counterexamples) there are zero violations, minimum slack $+0.768$. The index of c is the block length; since I is monotone increasing (Remark 15) the base value is the supremum along the dyadic chain, and numerically $c(\Delta; 8) = c(\Delta; 32)$ to 2×10^{-8} — the constants of v0.5 are unchanged.

Definition 2 (The split-causal sub-family). $\mathcal{F}_0^{\text{sc}}(m) \subset \mathcal{F}_0$ is the set of records whose block-1 cells read no block-2 source; in moment coordinates this is the *linear* condition

$$H[b_1, \text{idx}_2] = H[b_1, \text{idx}_1] \Sigma_{11}^{-1} \Sigma_{12},$$

idx_i the source indices of block i and $\Sigma = \Sigma_W$. Equivalently $X = 0$.

Corollary 3 ((H*) is a theorem on $\mathcal{F}_0^{\text{sc}}(m)$, with the same constants). On $\mathcal{F}_0^{\text{sc}}(m)$, $X = 0$ and hence $D_1 + D_2 \leq c(\Delta; m)$ — exactly the inequality (H*) asserts, now proved, with this document's own constants. Numerically: X vanishes to 1.0×10^{-14} on projected records and the minimum slack of $D_1 + D_2 \leq c(\Delta; m)$ over them is $+0.768$. Moreover $\mathcal{F}_0^{\text{sc}}(m)$ is a convex linear section of $\mathcal{D}$ (moment midpoints of split-causal records stay split-causal: residual 1.1×10^{-16} , midpoint $X \leq 5.1 \times 10^{-15}$), so the Lagrangian machinery of Theorem 2 restricted to the section certifies ϕ^{sc} two-sided exactly as it does for the diagonal section (Remark 5).

(n, Δ)	certified bracket for ϕ_n^{sc}	split-causality price $n \delta_n$ (bits)
(16, 0)	[0.5669667555, 0.5670144753]	0.004101 (≤ 0.004165)
(24, 0)	[0.5655717237, 0.5655854048]	0.004117 (≤ 0.004207)
(32, 0)	[0.5647601440, 0.5648697385]	0.004089 (≤ 0.004382)
(16, 1)	[0.5414842536, 0.5414911968]	0.010954 (≤ 0.011002)
(16, 2)	[0.5367829566, 0.5367912904]	0.014358 (≤ 0.014408)

Here $\delta_n = \phi_n^{\text{sc}} - \phi_n$ and the parenthesised figure is the certified (LB-to-UB) upper bound on the price. **The price is flat in n** at ≈ 0.0041 bits at $\Delta = 0$ across $n = 16, 24, 32$: split-causality costs a pure *boundary* constant, not a rate — which is what makes it a plausible object for a transfer argument in the first place.

Lemma 5 (Π — the split-causal retraction). *Let Π replace the block-1 rows of a record by their W^{b_1} -conditional means plus independent noise matched to restore $\text{Cov}(\hat{Y}^{b_1})$. Then Π maps $\mathcal{F}_0$ onto $\mathcal{F}_0^{\text{sc}}(m)$ and*

- (i) *preserves the per-cell distortion exactly (measured 1.1×10^{-16});*
- (ii) *preserves the entire block-1 joint law $(W^{b_1}, S^{b_1}, \hat{Y}^{b_1})$ exactly (6.7×10^{-16}), hence every own-window block-1 term;*
- (iii) *kills the cross-read, $X(\Pi R) \leq 2.1 \times 10^{-14}$;*
- (iv) *strictly improves generic records — over the pinned generic battery ($n \in \{12, 16, 20\}$, three styles, $\Delta \in \{0, 2\}$) the gain $n(L_a(\Pi R) - L_a(R))$ lies in $[-2.74, -0.31]$ bits (prover’s own battery: $[-3.11, -0.37]$);*
- (v) *costs only $\approx 5.0 \times 10^{-3}$ bits at optimizers (0.005052/0.005044/0.005039/0.005035 at $m = 8/12/16/24$, $\Delta = 0$).*

Π is what makes $\mathcal{F}_0^{\text{sc}}$ a retract and not merely a subset, and (i)–(iii) are what make the certified ϕ^{sc} brackets above meaningful. **What Π does not do is bound X** : at an optimizer its price is ≥ 0 automatically, so “price ≥ 0 ” is consistent with any X whatever — see Remark 17(b), where that route is closed. The measured price/ X ratio is 0.7960 at every n and every Δ .

Hypothesis 2 ((H^{**}) — the cross-block read at dyadic-chain optimizers; hard, load-bearing). *Fix Δ and a base window length n . At every optimizer of the dyadic chain $n \cdot 2^k$, $k = 0, 1, 2, \dots$, the cross-block read of the split into halves satisfies $X \leq \bar{X}$.*

Corollary 4 ((H^*) — what v0.5 assumed — is now a consequence). *Under (H^{**}) , Theorem 5 gives*

$$D_1 + D_2 \leq c(\Delta; n) + (2 - \mathbf{1}\{\Delta = 0\})\bar{X} =: c^*(\Delta; n),$$

which is (H^) with the constant enlarged by the repair term. (H^{**}) is strictly weaker and better margined: it constrains one scalar rather than a sum of $2m$ conditional mutual informations, the measured value at the $\Delta = 0$ optimizers is $X = 6.35 \times 10^{-3}$ against the largest admissible $\bar{X} = 32(\text{LB}(32, 0) - 0.5479448) - c(0) = 0.116930$ — $18\times$ headroom, against (H^*) ’s 0.076971 inside 0.4202858 ($5.5\times$) — it is verified monotone decreasing in m (Remark 13), and it is constructively bounded at any m by Π (Lemma 5).*

Theorem 6 (The split-causal transfer step; *unconditional*). For every m and every Δ ,

$$\phi_{2m}^{\text{sc}}(\Delta) \leq \phi_m(\Delta) :$$

concatenate two independent ε -optimal m -window records. The result is feasible on the $2m$ -window (the distortion constraint is the cell average) and is *split-causal at the midpoint* — its block-1 rows read no block-2 source whatever — and its value is at most $\phi_m + \varepsilon$ by the per-cell inequality of Theorem 4(i). Certified form (upper end of ϕ_{2m}^{sc} against lower end of ϕ_m): the inequality holds with certified slack $3.8/2.4/1.9 \times 10^{-3}$ at $(2m, m) = (16, 8)/(24, 12)/(32, 16)$. Applying Corollary 3 to the $2m$ -window $\mathcal{F}_0^{\text{sc}}$ -optimizer gives $\phi_m - \phi_{2m}^{\text{sc}} \leq c(\Delta; m)/2m$, whence

$$\phi_m(\Delta) - \phi_{2m}(\Delta) \leq \frac{c(\Delta; m) + p_{2m}}{2m}, \quad p_n := n(\phi_n^{\text{sc}} - \phi_n)$$

the *split-causality value gap* — the same quantity as the price column of the ϕ^{sc} table above.

Hypothesis 3 ((H***) — the split-causality value gap along the chain). $p_m \leq \bar{p}$ for every m in the dyadic chain.

Remark 11 (Three hypotheses, and why (H***) is the one to prefer). Each of (H*), (H**), (H***) suffices for Theorem 7, and they are ordered by how little they assume and — more importantly — by how *checkable* they are. (H**) implies (H*) with an enlarged constant, by Theorem 5 (Corollary 4); it replaces a bound on a sum of $2m$ conditional mutual informations by a bound on one scalar. (H***) replaces that scalar by another, p_m , which reaches the same conclusion through Theorem 6 rather than through Theorem 5 — and p_m **is a gap between two convex programs, so Theorem 2 certifies it two-sided**, whereas X is only ever *measured* at a computed optimizer. That is the decisive difference: a hypothesis about p is falsifiable by certificate at any m one is willing to pay for. Both bars are the same number, $\bar{p}$, $\bar{X} < 32(\text{LB}(32, 0) - 0.5479448) - c(0) = 0.116930$; the certified instances — each the LB-to-UB upper end, so each an honest *bound* on p_n rather than an optimizer reading — are $p_{16} \leq 0.004165$, $p_{24} \leq 0.004207$, $p_{32} \leq 0.004382$: flat in n , and $27\times$ inside the bar, against X 's $18\times$.

Remark 12 (The “ $\leq \kappa$ per side” refutation, *refined* — what it refutes is split-causality). The natural lemma — that each side's boundary charge is at most the AR(1) boundary information κ , uniformly over the family — is *false*, and its sharpened form $D_i \leq \kappa - I_m$ is false too. Explicit D -feasible records in $\mathcal{F}_0$ (no U -coupling whatsoever; the whole budget respected) spend block-1 rows on copies of block-2 cells:

- two V -copy rows give $D_2 = 0.4563 > 0.4203 = \kappa - I_8$: the *sharpened* constant is exceeded by a mild, feasible perturbation;
- three Y -copy rows give $D_2 = 18.93$ bits $\approx 26\kappa$ at distortion 0.2986: the charge is not bounded by κ at all.

Nothing stops the construction from being repeated: the worst case over the family is $\Theta(m)$ in the block length. **No universal constant exists** — that statement stands unchanged.

What has changed is the diagnosis. These are precisely the records with enormous cross-block read: the harness's own witnesses of the same class measure $X = 18.46$ and $X = 29.27$ bits (against 6.3×10^{-3} at the optimizers), and at the two- V -copy witness the bound of Lemma 2 is *tight to*

5.7×10^{-7} . The refutation is exactly the failure of split-causality, and Theorem 5 absorbs it into the term $(2 - \mathbf{1}\{\Delta = 0\})X$ rather than being defeated by it. Hence the v0.5 blanket rule (“this document never prints $\leq \kappa$ per side as an $\mathcal{F}_0$ lemma”) is *refined*, not repealed: $D_1 + D_2 \leq c(\Delta; m)$ is a *theorem* on $\mathcal{F}_0^{\text{sc}}(m)$ (Corollary 3) and is *false* on $\mathcal{F}_0$; neither form may be printed without its scope. What no longer follows is the v0.5 sentence “no proof of the reverse direction can avoid using optimizer structure”: the *inequality* of Theorem 5 avoids it entirely; what still needs optimizer structure is a bound on X (Remark 17).

The two constants above are the verifier’s witnesses; they are not in the repository, so the harness pins its own analytic records of the same class and measures $D_2 = 0.7749$ for two V -copy rows (distortion 0.2706, $L_a(0) = 4.219$) and $D_2 = 13.94$ bits $\approx 19\kappa$ for three Y -copy rows (distortion 0.2754, $L_a(0) = 4.740$). Same refutation, independent witnesses; the refutation itself is a gated fact of the harness, not a remark. Note the L_a values: these records are nowhere near optimal — more than seven times $\phi_{16} = 0.5668$ — which is why they never threatened the transfer bound in practice, only its proof.

Remark 13 (The monotone $m \times \Delta$ table (the “ $m \in \{8, 16\}$ only” caveat, retired)). v0.5 recorded the optimizer verifications at $m \in \{8, 16\}$ only and flagged the gap explicitly. The table is now $m \in \{8, 12, 16, 24\}$ at $\Delta = 0$ and $m \in \{8, 12, 16\}$ at $\Delta = 1, 2$ (cold-start certified full-space optimizers at $n = 2m$), and *every* quantity of the repair is monotone *decreasing* in m :

Δ	m	$D_1 + D_2$	$I(E; T^{b_2})$	X
0	8/12/16/24	0.076971/0.076934/0.076915/0.076896	0.543069/0.543059/0.543054/0.543049	6.3468/6.336
1	8/12/16	0.090533/0.090487/0.090463	0.569542/0.569504/0.569485	1.8013/1.798
2	8/12/16	0.099029/0.098964/0.098931	0.579319/0.579256/0.579225	2.3894/2.384

Monotonicity is what the caveat was about: *the base value dominates the whole chain*, so a hypothesis verified at the base m is not being extrapolated upward along the doubling chain. *Scope of the netting*: the $\Delta = 0$ row is reproduced end-to-end and gated (strict decrease at every step, each step $\geq 10^{-6}$ against a smallest measured step of 4.96×10^{-6}); the $\Delta = 1, 2$ rows are the prover’s, carried on its authority and not re-netted, except that the $m = 8$ optimizers at $\Delta = 1, 2$ are recomputed for Theorem 3. Two-point $1/m$ limits at $\Delta = 0$: $X_\infty \approx 6.31 \times 10^{-3}$, $(D_1 + D_2)_\infty \approx 0.076859$, $I(E; T^{b_2})_\infty \approx 0.54304$; the Π -price of Lemma 5(v) is monotone too, limit $\approx 5.03 \times 10^{-3}$.

Two precision notes on the values v0.5 recorded at $m = 16$. (a) The recorded $D_2 = 0.076926$ and $I(E; T^{b_2}) = 0.543057$ came from a *warm-polish* endpoint sitting at $d - D = +6.4 \times 10^{-5}$, i.e. *distortion-infeasible*; the feasible cold-start certified values are 0.0769149 and 0.543054. (b) The difference, 1.1×10^{-5} resp. 3×10^{-6} , is more than two orders inside the $5 \times 10^{-3} / 2 \times 10^{-2}$ reproduction bands the 080 harness gates these two quantities at — so this is a *precision note*, *not a bar change*, and the sealed 080 verdicts are untouched. The table above quotes the feasible cold-start values throughout.

Theorem 7 (T^* v2 — boundary-charged reverse direction and the two-sided bound, in *two* forms). Fix Δ and a base n , and write $c^*(\Delta; n) = c(\Delta; n) + (2 - \mathbf{1}\{\Delta = 0\})\bar{X}$.

(ii-a) *Family form, conditional on Hypothesis 2 along the chain $n \cdot 2^k$* : $\phi_m(\Delta) - \phi_{2m}(\Delta) \leq \frac{c^*(\Delta; n)}{2m}$
for every $m = n \cdot 2^k$, and, summing the geometric telescope, $0 \leq \phi_n(\Delta) - L^\infty(\Delta) \leq \frac{c^*(\Delta; n)}{n}$,
the lower half being Corollary 2 and *unconditional*.

(ii-b) *Split-causal form*, unconditional: the same two inequalities hold for the sub-family value $\phi_n^{\text{sc}}(\Delta) = \inf_{\mathcal{F}_0^{\text{sc}}} L_a$ with the constant $c(\Delta; n)$ itself and *without any of* (H*), (H**), (H***), because on $\mathcal{F}_0^{\text{sc}}$ the charge bound is Corollary 3. (It remains under Hypothesis 1, like everything else in this document, and this scoping is never to be restated as freedom from *every* hypothesis — for the plateau in particular, see Remark 20.)

Scope of (ii-b). It bounds $\phi_n^{\text{sc}} - L^{\text{sc}, \infty}$, not $\phi_n - L^\infty$; the two coincide iff the split-causality price $\delta_n = \phi_n^{\text{sc}} - \phi_n$ is $o(1/n)$ in the relevant sense. Measured, $n\delta_n$ is *flat* at ≈ 0.0041 bits over $n = 16, 24, 32$ at $\Delta = 0$ (so $\delta_n = \Theta(1/n)$, of the same order as the transfer term itself, not smaller). That is evidence, not a proof, and (ii-b) is therefore *not* a route to the family statement — it is the unconditional theorem about the sub-family, and the honest statement of what the repair buys without (H**).

Exact superadditivity is refuted, so (ii) cannot be sharpened to $c = 0$: the certified anchors are strictly decreasing, $\phi_8 > \phi_{12} > \phi_{16} > \phi_{24} > \phi_{32}$, with the certified transfer anchors $\phi_8(0) \in [0.570790938, 0.5707933842]$ and $\phi_{12}(0) \in [0.568030932, 0.5681028134]$. The bound is not tight either: the measured drift $n(\phi_n - L^\infty) \approx 0.0645$ at $n = 32$ sits $6.5\times$ inside $c(0) = 0.4203$.

Remark 14 (The per-block feasibility step of (ii) — a separate hole, and its repair). The restriction step behind (ii) needs each half of the $2m$ -optimizer to be a *feasible* m -window record, i.e. $\text{own}_i \geq m\phi_m(D)$, and v0.5 tacitly assumed each block distortion is $\leq D$. **It is not.** The $2m$ -optimizer's block distortions *straddle* D — $d_1 = 0.299486$, $d_2 = 0.300514$ at $n = 16$; $0.299658/0.300342$ at $n = 24$; $0.299744/0.300256$ at $n = 32$ — with the *mean* exactly D (to 5×10^{-8}). The correct step is

$$\text{own}_1 + \text{own}_2 \geq m[\phi_m(d_1) + \phi_m(d_2)] \geq 2m\phi_m(D),$$

the second inequality by *convexity of the value function* $D \mapsto \phi_m(D)$, which Theorem 2 supplies for free (jointly convex objective, affine distortion constraint $\Rightarrow$ convex optimal value in the right-hand side). Certified on the pinned grid $D \in \{0.26, 0.28, 0.30, 0.32, 0.34\}$ at $m = 8$, $\Delta = 0$: at each interior node the certified *upper* end sits below the chord of the two certified *lower* ends, slack $1.90/1.64/1.47 \times 10^{-3}$. Ignoring the straddle would have cost $m\mu(d_2 - D) = 8 \times 2.2206 \times 5.14 \times 10^{-4} = 9.13 \times 10^{-3}$ bits at $n = 16$ (μ the distortion multiplier at the optimizer) — *larger than the entire repair term* $\bar{X}$, so the hole was not cosmetic. It is independent of (H*) and of everything else in this section.

Remark 15 (The constants, and their n -uniformity). $I_n = I(S^{b_1}; S'^{1.. \Delta+1})$ is *monotone increasing in* n , so $c(\Delta; n) = (2 - \mathbf{1}\{\Delta = 0\})\kappa - I_n$ is the supremum of the per-step constants along the whole dyadic chain issuing from base n — which is precisely what the telescope in Theorem 7(iii) requires, and why the constant must be built from I_n and not from I_∞ . Numerically I_n is converged to I_{32} by $n = 16$ (to $< 10^{-12}$), and

$$I_{32} = 0.3166798 / 0.3482018 / 0.3520686 \quad \text{at } \Delta = 0/1/2,$$

each equal to I_∞ to below 10^{-15} . Hence

$$c(0) \leq 0.42029, \quad \boxed{c(1) \leq 1.1258}, \quad c(2) \leq 1.1219.$$

Erratum: the transfer note recorded $c(1) \leq 1.1257$; the computed value is 1.1257294, so 1.1257 is false in the fifth decimal. Use 1.1258.

The repaired constants (v0.6). With $c^*(\Delta; n) = c(\Delta; n) + (2 - \mathbf{1}\{\Delta = 0\})\bar{X}$ and $\bar{X}$ at the base- m measured values of Remark 13, $c^*(0) = 0.4266326$, $c^*(1) = 1.161755$, $c^*(2) = 1.169651$ — the $\Delta = 0$ constant grows by 1.5%, the others by 3–4%, because X itself grows with Δ and enters doubled there. Only $c^*(0)$ is used in Corollary 5; $\Delta = 1, 2$ remain $O(1/n)$ -model-conditional for the reason given there, and the repair does not change that.

Remark 16 (The zero-claim and the marginalization step are $\mathcal{F}_0$ -only). Two steps carry the whole cancellation. (a) The *zero-claim* $I(E; S'_j \mid \text{pfx}_j, T^{b_2}) = 0$ makes the interleaved chain rule cancel the record–reference coupling exactly (residual $\leq 2 \times 10^{-14}$ over pinned $\mathcal{F}_0$ records). (b) The *marginalization* step replaces the block-2 sub-record by an effective 8-window record with the same law (deviation 2×10^{-16} ; the effective noise is independent of both W^{b_2} and U^{b_2}). **Both are tagged $\mathcal{F}_0$ -only.** They fail outside the family, and the failure is not a technicality: with block-1 rows coupled to U^{b_2} the zero-claim is violated by 0.2223 bits, and with the coupling confined to *block-2 rows only* it is still violated by 0.0950 bits — in the latter case the violation is induced purely through the conditioning, with no block-1 record touching U at all. So U -independence is load-bearing here in the same way it is load-bearing in Theorem 1 (Remark 3), and for a third time.

Corollary 5 (The $\Delta = 0$ plateau over the causal-spectral allocation — recomputed with the repaired constant. **HISTORICAL at v0.7). Superseded in full by Corollary 7.** *The arithmetic of this corollary — both the (H^{**}) number 0.5514005 and the $(D)+(F)$ number 0.5495664, and both their margins — is retained only as history. It is not the plateau statement of this document; §6 derives the plateau with none of (H^*) , (H^{**}) , (H^{***}) , (D) , (F) or the boundary charge, at margins larger by a factor of 4.3 at $\Delta = 0$ and at two further lags this route cannot reach at all. Nothing downstream may cite the numbers below.*

Assume Hypothesis 2 at $\Delta = 0$ with base $n = 32$, with $\bar{X}$ taken at the measured value $X = 6.346782 \times 10^{-3}$. With the sealed 079 certified lower end $\text{LB}(32, 0) = 0.5647327869$ (harness `go14_convexity.py`, entry `s5_32.0`) and $c^*(0) = c(0) + X = 0.4202858 + 0.0063468 = 0.4266326$,

$$L^\infty(0) \geq 0.5647327869 - \frac{0.4266326}{32} = 0.5514005 > 0.5479448,$$

the right-hand side being the causal-spectral allocation at $\Delta = 0$ (Remark 9). The margin is $+3.456 \times 10^{-3}$: $203 \times$ the 1.7×10^{-5} bracket width quoted for $(32, 0)$ in the anchor table of §3.2, and $376 \times$ the 9.18×10^{-6} width the sealed harness actually achieved. This is a statement about the limit itself, not about the $O(1/n)$ finite-window model. (At v0.5, under (H^) , the same arithmetic gave 0.5515989 at $215 \times / 398 \times$; the repair costs 1.98×10^{-4} of margin and buys a hypothesis with $18 \times$ headroom instead of $5.5 \times$.)*

Under the decay route instead. Replacing (H^{**}) by the two model-scale hypotheses (D) and (F) of Remark 17 gives the proved envelope $\bar{X} = 0.065038$, hence $c^*(0) = 0.485324$ and

$$L^\infty(0) \geq 0.5647327869 - \frac{0.485324}{32} = 0.5495664 > 0.5479448,$$

*margin $+1.622 \times 10^{-3}$ ($95 \times$ the quoted bracket, $177 \times$ the sealed width). Which hypothesis buys which margin is therefore explicit: (H^{**}) — one measured scalar — buys 3.456×10^{-3} ; $(D)+(F)$ — two structural, m -uniform statements with a proved conversion — buys 1.622×10^{-3} . Neither is discharged.*

The margin is not slack in the constant, and the base matters. *Running the same arithmetic from base $n = 24$ gives $0.5654101216 - 0.4266326/24 = 0.5476338 < 0.5479448$: it fails by 3.11×10^{-4} (and by 2.757×10^{-3} under (D)+(F)) — so the enlarged constant makes the no-slack claim stronger than at $v0.5$, where the base-24 shortfall was 4.7×10^{-5} . The hypothesis is genuinely load-bearing, and so is the choice of base window. $\Delta = 1$ and $\Delta = 2$ remain $O(1/n)$ -model-conditional: their constants are $2.7\times$ larger and their plateaus $3.6/18\times$ smaller, so a certified anchor at $n \approx 260$ resp. $n \approx 1320$, or a sharper constant, would be needed.*

Remark 17 (The reduction, executed — and the obstruction, named). $v0.5$ recorded a *repair path*. It has since been walked, and it ends at a named theorem rather than at a vague hope. Three things are now on record.

(1) *The conversion is proved and is m -uniform by construction.* Exactly, $X = \frac{1}{2} \log_2 \det(I + N_{11}^{-1} A^{12} \Sigma_{2|1} (A^{12})^\top)$ with N the record noise covariance and A^{12} the cross-block kernel. Under

(D) *kernel-decay envelopes:* $|A_v[t, s]| \leq C_v \lambda^{|t-s|}$, $|A_y[t, s]| \leq C_y \lambda^{|t-s|}$ with $\lambda = \lambda_s = 0.354554$, $C_v = 0.1980$, $C_y = 0.1631$ (fitted, and identical to three digits at $n = 24/32/48$), and

(F) a uniform noise floor $\lambda_{\min}(N) \geq \nu = 0.1495$,

one gets a closed-form bound on X whose sums run over infinite half-lines and which is therefore *independent of m* ; the conditional covariance $\Sigma_{2|1}$ is used exactly rather than enveloped (its $-a^{j+k+2}$ boundary term is worth $2.6\times$ and is what makes the bound clear the bar). Audited link by link at real optimizers the total looseness is $10.3\times$ ($\log \det \rightarrow \text{tr } 1.004\times$, $N_{11} \succeq \nu I$ $1.22\times$, the enveloped Schur step $8.4\times$). **Result:** $\bar{X} = 0.065038$ **against the admissible** 0.116930 — **it clears with $1.80\times$ margin, uniformly in m** (free- λ optimisation gives 0.0648 ; insensitive to the envelope-fit window). So (H**) is no longer a bare hypothesis but a *consequence* of (D) and (F), at the arithmetic cost recorded in Corollary 5.

(2) *The spectral premise (D) is not the obstruction — it holds numerically, uniformly in m .* At $\Delta = 0$, $m = 8/12/16/24$: θ moves only over 3.0785 – 3.0835 , $\text{cond } N$ over 1.4453 – 1.4492 , $\text{cond } M_Q$ over 2.197 – 2.217 , $\lambda_{\min}(J) = 0.2686$ flat, the record pivots $\sigma_t \in [0.3753, 0.4588]$, and $\|A^{12}\|_F$ is *flat in m* ($5.4823 \rightarrow 5.4722 \times 10^{-2}$; $\|A^{12}\|_F / \|A^{11}\|_F = 3.44\%$). The measured kernel is the same Toeplitz kernel plus non-moving boundary corrections at every n , and its per-lag ratios *decrease* with lag ($0.284 \rightarrow 0.200$ for A_y , $0.377 \rightarrow 0.279$ for A_v), so a single geometric envelope at λ_s is conservative.

(3) *The obstruction is **circularity of the operator**, and it is a genuine research step.* By Theorem 3 the only operator whose inverse produces the optimizer is $N = (\theta I + V \Sigma_s^{-1} V^\top)^{-1}$, and $M := V \Sigma_s^{-1} V^\top = N^{-1} - \theta I$ *exactly*: “ M decays exponentially” and “ N decays exponentially” are the same statement, so a Demko/Jaffard inverse-closedness theorem applied here *returns its own hypothesis*. The model’s own decay ($\Sigma_S, \Sigma_W, \Sigma_{W|S}$ all exactly $a^{|t-s|}$) enters the KKT system only through the affine data, never through the operator being inverted. What would inject it is a *fixed-point/bootstrap* argument in a Jaffard (or Gröchenig–Klotz weighted) Banach algebra: show that the KKT map sends a ball $\{\|A\|_{\mathcal{A}_\lambda} \leq R\}$ into itself and contracts. A second, independent piece is missing too: (F), the uniform lower bound $\lambda_{\min}(N) \geq \nu$. Theorem 3 gives the *upper* side ($N \preceq I/\theta$, proved) but not the lower; every attempted route closes on itself ($\lambda_{\max}(M) \leq 1/\lambda_{\min}(M_Q)$ with $M_Q \succeq N$ gives only $1/\nu \leq \theta + 1/\nu$), and coercivity fails because a record that does not use a direction pays nothing there.

Routes closed, so that they are not re-attempted. (a) KKT zeroing cannot work: since $\Sigma_W^{-1} \Sigma_{WY} = [0; I]$, the distortion gradient *vanishes identically* on the cross-read coordinates (Theorem 3), so stationarity there carries no multiplier — the cross-read is a *distortion-free* direction on which the

objective is stationary, and the optimizer chooses $X > 0$ because it strictly lowers L_a . KKT yields an equation, not a bound. (b) The Π -retraction cannot bound X either: at an optimizer the Π -price is ≥ 0 automatically, and the measured price/ X ratio is 0.7960 at every n and every Δ , so “price ≥ 0 ” is consistent with *any* X . (c) m -induction restates the same missing estimate: X is monotone in m (Remark 13) but the step compares the m - and $2m$ -optimizers. *The next route on record is different in kind*: solve the stationary system (K1)–(K3) in the spectral domain with the GO-13 machinery, where it becomes per-frequency scalar equations — rational symbols with poles off the unit circle would give Toeplitz coefficients decaying at exactly the modulus of the nearest pole, which the numerics place at $\lambda_s = a(1 - K) = 0.354554$, GO-14’s own netted constant — and then transfer stationary $\rightarrow$ finite-window by a stability estimate from the strong convexity of Theorem 2.

What this does not do. Corollary 5 cannot drop its hypothesis. It trades (H**) for (D)+(F); the plateau survives either way, at $+3.456 \times 10^{-3}$ resp. $+1.622 \times 10^{-3}$, and the base-24 control fails either way.

5.2 The within-class ladder, certified (the Remark 5 upgrade, executed)

The diagonal class is a convex *linear* section of $\mathcal{D}$ (Remark 5), so the Lagrangian machinery of Theorem 2 restricted to the section — with a section moment box $R_{\text{box}} = 73/95/136$ at $n = 24/32/48$, verified attained and hence tight — certifies the ladder two-sided. All nine ladder cells and the three block values are now certified *class* values (per cell the tighter of the two independent end-to-end computations):

(n, Δ)	certified within-class bracket	width
(24, 4)	[0.5333160113, 0.5333170450]	1.0×10^{-6}
(24, 5)	[0.5332969026, 0.5332971560]	2.5×10^{-7}
(24, 6)	[0.5332945088, 0.5332947270]	2.2×10^{-7}
(32, 4)	[0.5324808853, 0.5324814627]	5.8×10^{-7}
(32, 5)	[0.5324606031, 0.5324610143]	4.1×10^{-7}
(32, 6)	[0.5324578912, 0.5324584264]	5.4×10^{-7}
(48, 4)	[0.5316458819, 0.5316462767]	3.9×10^{-7}
(48, 5)	[0.5316244073, 0.5316250208]	6.1×10^{-7}
(48, 6)	[0.5316217661, 0.5316233575]	1.6×10^{-6}
block ₂₄	[0.5332936495, 0.5332949450]	1.3×10^{-6}
block ₃₂	[0.5324577110, 0.5324581027]	3.9×10^{-7}
block ₄₈	[0.5316214213, 0.5316220475]	6.3×10^{-7}

The (48, 6) cell is the loosest (its feasible-projection correction is 1.1×10^{-6} , still inside the bracket), which is the same cell the raw-anchor demotion of Remark 8 concerns.

What this does *not* certify. The excess $E_\infty(\Delta)$ is a *two-point* $1/n$ *extrapolation* of these class values against the block values, so the re-assembled E_∞ intervals are $1/n$ -*model-conditional*, *not certificates*, and are tagged as such: $E_\infty(4) \in [2.50, 2.75] \times 10^{-5}$, confirming the 2.67×10^{-5} of Conjecture 1 in bar; $E_\infty(5) \in [1.7, 4.8] \times 10^{-6}$, sign resolved positive by the tighter re-assembly and *unresolved* by the looser one; and $E_\infty(6) \in [-3.2, 1.5] \times 10^{-6}$, whose bracket *contains zero* at every achievable width. The last of these independently re-ratifies the R-IND-5 demotion of $E_\infty(6)$ to the raw anchor $L_a^{\text{diag}}(48, 6) = 0.5316222730$ (Remark 8). No claim is made here about how

many optimizer endpoints were true class minima; the transfer note’s “4/9 endpoints” remark is *withdrawn* as unverifiable from the record.

6 The plateau: Collapse, periodization, shift-averaging, and a two-sided certificate for Ψ

Status: spectral prover + Ψ -bracket prover + two independent R-IND-5 passes (the bypass pass, ten restatements; the certificate pass, ten further restatements R11–R20 — fresh context, own spectra, own solver, own gradient, own box, own dual, structurally unrelated cell-argmin fixed point rather than the K1–K3 filter recursion). All twenty restatements are in force in this revision. Harness `experiments/go14_plateau.py`; registration 082 pending — **nothing in this section is sealed**.

The stationary frame. Write $R_u := \hat{Y}_{u-\Delta-1}$. In this shifted frame the interleaved order of the coordinate becomes the plain simultaneous order of (R, S) , and Δ **enters only as the phase** $z^{\Delta+1}$ **on the record’s cross spectra** — that is the entire content of the lag coordinate at stationarity. Per frequency put

$$h(\omega) = (\Phi_{RV}, \Phi_{RY}), \quad \Gamma(\omega) = \Phi_R, \quad P = \Sigma_W(\omega)^{-1}, \quad Q = \text{diag}(1/f_S, 0),$$

$n(\omega) = \Gamma - hPh^* \geq 0$ the cone, $M_Q = \Gamma - hQh^*$, and $R = P - Q$. The reduction of Q to the rank-one diagonal $\text{diag}(1/f_S, 0)$ is *exact* and is Theorem 3’s first structural fact $\Sigma_W^{-1}\Sigma_{WS} = e_1$ (re-verified to 1.5×10^{-15}); $R \succ 0$ in closed form, $\det R = (1/f_V - 1/f_S)/s_N^2 > 0$ (verified to 5.3×10^{-15} ; $\lambda_{\min}(R) \geq 3.17 \times 10^{-3}$ on the grid). Write $\Psi(D; \Delta)$ for the value of the stationary program in these coordinates and $\alpha = 1/(2 \ln 2)$.

6.1 The Collapse identity

Theorem 8 (Collapse — exact, unconditional). For every n , every nondecreasing access schedule and every record of $\mathcal{F}_0$,

$$\boxed{2 \ln 2 \, n L_a = \sum_{t=1}^n \ln \sigma_t - \ln \det N} \quad \sigma_t = \text{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1}),$$

$N = \text{Cov}(Z)$. *Proof.* Conditioning on W reduces $\hat{Y}_t$ to Z_t and $S^{se(t)}$ to $U^{se(t)} \perp Z$, so every denominator of the coordinate is $\text{Var}(Z_t \mid Z^{t-1})$ and the denominator sum telescopes to $\ln \det N$. $\square$
Certificates: relative residual $\leq 1.2 \times 10^{-14}$ at all ten certified optimizers ($n = 16/24/32/48 \times \Delta = 0/1/2$); independent R-IND-5 evaluator 5.6×10^{-15} over 84 cells, 1.7×10^{-15} on random non-staircase nondecreasing schedules, 8.2×10^{-16} at the edges; harness 5.65×10^{-15} and 1.73×10^{-15} against a 10^{-12} gate.

The identity re-derives (K1)–(K3) in five lines and makes step (A) below a one-liner.

Remark 18 (R6 — what Collapse actually needs, and the non-conflation rule). Collapse does **not** require U-independence. It requires exactly Δ -lag-causal U -coupling, $A_u[t, s] = 0$ for $s > t + \Delta$; Hypothesis 1 ($A_u = 0$) is the special case. Counter-values on record: at $\Delta = 0$ a dense A_u puts the identity 0.2018 bits off and a strictly anticausal A_u 0.3329 bits off, while a Δ -lag-causal A_u is *exactly* 0 (harness: ≥ 0.1485 bits and 6.7×10^{-16} respectively, gated in both directions). **The**

non-conflation rule of Remark 3 still applies: this extension is a statement about the *record-space* coordinate and does *not* extend the moment-form Theorem 1, whose measured residuals at Δ -causal couplings are 2.4–3.5 bits. The two extension results must not be conflated.

6.2 Step (A): periodization

Hypothesis 4 ((IC) — independent noise copies; a hypothesis of the *construction*, load-bearing). *In the periodization of Theorem 9 the noise blocks of distinct translates are independent: the tiled noise covariance is block diagonal, $N_{\text{tiled}} = \bigoplus_b N$.*

Theorem 9 (A — periodization; Ω_n/n subadditive). Repeat any n -record along $\mathbb{Z}$ under Hypothesis 4. Then (i) the per-symbol distortion is *exactly* preserved; (ii) $\ln \det N$ is *exactly* additive; (iii) every σ_t decreases, because the block- b translate’s conditioning set is a genuine *subset* of the tiled cell’s — $bn + \min(i + \Delta, n) = \min(bn + i + \Delta, bn + n) \leq \min(bn + i + \Delta, M)$. By Theorem 8 the tiled per-symbol value is therefore no larger, so

$$L^{\text{process}}(D) \leq \phi_n(\Delta) \quad \text{for every } n.$$

This is Theorem 4 in one line instead of a page, and it carries *no* boundary charge because it runs the easy way. *Certificates:* distortion preserved $\leq 4.4 \times 10^{-16}$, $\ln \det N$ additive $\leq 7.1 \times 10^{-15}$, and every σ_t decreases over 1,200 (cell $\times$ tiling) pairs with *zero* violations — the suspected failure mode (a merely *different* rather than *larger* conditioning set) does not occur, for the displayed reason. Measured boundary charge 0.04205/ n , flat to five digits at the certified optimizers.

Remark 19 (R7 — (IC) is load-bearing, not decorative). Correlating the noise copies *breaks* Theorem 9 outright. At cross-block noise correlation 0.3/0.6/0.9 the tiled per-symbol value is 0.7987/0.9218/1.3977 against $\phi_5 = 0.7757$; the harness’s own witnesses exceed the untiled value by +0.0290/ +0.1605/ +0.6463 on a random $n = 5$ record and by +0.0319/ +0.1684/ +0.6836 over ϕ_8 at the cold-start (8,0) optimizer. So (IC) must be stated as a hypothesis of the construction and never dropped in restatement.

6.3 Step (B): shift-averaging

Lemma 6 ((R1) — convexity of the process-rate functional on the cyclostationary class). *Let F_T denote the total-rate functional on the window of length T in process-moment coordinates. Then (i) F_T is convex on the linear section cut out by the cyclostationary class; (ii) on records whose spectra are bounded above and bounded away from 0, F_T/T converges pointwise to the process rate; and (iii) a pointwise limit of convex functions is convex. Hence the process-rate functional is convex on the cyclostationary class. **This lemma is stated here as a lemma, not cited as a corollary of Theorem 2:** Theorem 2 is a finite-window statement and the passage to the process rate is exactly the content of (ii)–(iii). Status: this is the single remaining load-bearing hypothesis of the plateau (together with Hypothesis 4), and it is a pointwise-limit-of-convex-functions statement — a much softer object than (H^*) , (H^{**}) , (H^{***}) , (D) or (F) . Numerics: 69 Jensen points and 40 curvature lines on the cyclostationary class (the class the argument actually touches, not merely the stationary one), zero violations.*

Theorem 10 (B — shift-averaging produces an explicit stationary record). Let x be a period- n record on $\mathbb{Z}$ and let $\bar{x}$ be the shift-average of its moment functions over the n shifts ($\bar{A}$ the diagonal average of A , $\bar{\Gamma}$ the period average, the noise spectrum explicit). Then $\bar{x}$ is

- (i) *in the cone*: the residual noise spectrum $\bar{n}(\omega)$ is nonnegative, with a named reason — the kernel-dispersion term is a variance, hence PSD, and it is exactly what makes the inequality strict;
- (ii) *stationary*, and *realisable*: the record is constructed, not merely asserted;
- (iii) of *identical per-symbol distortion*; and
- (iv) of *no larger rate*, by Lemma 6: $\text{value}(\bar{x}) \leq \overline{\text{value}(x)} = \text{value}(x)$.

Wording. Step (B) *delivers an explicit stationary record of no larger rate and identical per-symbol distortion*. It does *not* say that the process infimum is attained: attainment is never invoked, the infimum comparison follows by definition, and the lower-semicontinuity “soft spot” recorded in the probe is a red herring. *Certificates*: executed at the certified optimizers, the constructed stationary records evaluate to 0.5631757405/0.5629264128/0.5628390620 at $n = 8/12/16$, $\Delta = 0$ (harness, cold-start: 0.5631757430/0.5629263942/0.5628390182), distortion preserved to machine precision, converging from *above* like C/n^2 ; 8/8 adversarial edge cases (near-deterministic noise, sign-alternating kernels, rank-2 noise, strictly anticausal A_v) hold with cone margin ≥ 0.2 . There is no schedule leak: on $\mathbb{Z}$ the schedule *is* shift-invariant, and the finite- n truncation appears only on the $\leq$ side of Theorem 9. At $\Delta = 1, 2$ the same construction gives 0.5364993215 and 0.5311645087, reproduced by the harness to ten digits.

Corollary 6 (The chain, and its direction). $\phi_n \geq \text{rate}(\text{periodize}) \geq \text{rate}(\text{shift-average}) \geq \Psi(D; \Delta)$, whence

$$\boxed{L^\infty(\Delta) \geq \Psi(D; \Delta)}$$

under Hypothesis 4 and Lemma 6. **Every step is an inequality the same way and each produces an explicit feasible object, so the chain is one-directional**: it cannot yield the reverse inequality. The converse would need an unwritten truncation/achievability lemma — and the stationary optimum is not rational at any finite order, so the FIR-density step is not free; it is a boundary-charge argument again, in the other direction.

Remark 20 (What the chain does *not* give). Corollary 6 is a *lower* bound. The reverse inequality is *not* available from anything in this document, and no statement that the process limit coincides with Ψ may be printed. The permitted headline at $\Delta = 0$ is the bracket

$$L^\infty(0) \in [0.5627265, 0.5647420],$$

of width 2.0×10^{-3} : the lower end from Corollary 6, the upper end the certified $n = 32$ anchor of §3.2 through Corollary 2. Equally, the plateau is *not* hypothesis-free: it removes (H*), (H**), (H***), (D), (F) and the boundary charge — a real and large advance — and replaces them with Lemma 6 and Hypothesis 4. It remains under Hypothesis 1 throughout.

Remark 21 (Where the $O(1/n)$ drift lives). At $\Delta = 0$ the drift splits exactly along the two steps: $n(\phi_n - \Psi) = 0.0645 = 0.0420$ (periodization / boundary) + 0.0225 (cyclostationarity) + residual. The entire (H*)/(H**)/(H***) machinery was fighting only the first 65%.

6.4 The two ingredients of the certificate

Lemma 7 (S — evaluation of the leak term). *Under Theorem 8 and bivariate Wiener–Masani/Szegő, the stationary rate is*

$$\text{rate} = \alpha[\langle \ln M_Q \rangle - \langle \ln n \rangle] + \alpha[\langle \ln f_S \rangle - \ln s], \quad s = \text{Var}(S_u | S^{u-1}, R^u),$$

with the closed form $\langle \ln f_S \rangle = \ln(\tau^2 a / \lambda_s) = -0.1025391956$. **Scope (non-degeneracy).** The Szegő step needs the S -side spectrum bounded away from 0 and ∞ ; here $f_S \in [0.5111, 9.400]$, so it cannot degenerate — but the hypothesis is part of the lemma and must travel with it. Certificates: identity residuals $-2.2 \times 10^{-16} / -4.4 \times 10^{-16} / -1.1 \times 10^{-15}$ at the three certified points; the closed form agrees with the grid to 2.9×10^{-16} across $N_f = 256$ –16384. **Evaluator caveat.** On records with a deep narrow notch in the noise spectrum the finite-lag evaluator for σ and s converges very slowly (residual 2.4×10^{-4} at $P = 50$, still 1.5×10^{-4} at $P = 1600$). This is a property of the evaluator, not a failure of the lemma (which is P -independent to 4.4×10^{-16} at the anchors), and it does not touch the certificate, whose leak term is an exact integral of frozen filters. By-product: $\text{rate}_{\text{block+leak}} \leq \text{rate}_{\text{true}} \leq \text{rate}_{\text{collapse}}$ there, so the empirical gates used the aggressive side.

Lemma 8 (C-stat — per-frequency joint convexity). *Fix a frequency. If $Q \succeq 0$ and $R = P - Q \succeq 0$, then*

$$(h, \Gamma) \mapsto \ln(\Gamma - hQh^*) - \ln(\Gamma - hPh^*) = -\ln(1 - Z), \quad Z = \frac{hRh^*}{M_Q},$$

is jointly convex on the cone. Proof. $\varphi(h, t) = hRh^*/t$ is jointly convex on $t > 0$ (quadratic-over-linear, $R \succeq 0$) and nonincreasing in t ; $M_Q(h, \Gamma) = \Gamma - hQh^*$ is jointly concave ($Q \succeq 0$); h enters φ ’s first slot affinely. The composition rule for a convex, coordinatewise-monotone outer function gives Z convex; and $u \mapsto -\ln(1 - u)$ is convex increasing on $u < 1$. $\square$

Hypotheses used are exactly $Q \succeq 0$ and $R = P - Q \succeq 0$; both are load-bearing (R12). No matrix lift is needed and the 074 lemma (GO-13 Theorem 5, the block Schur lift of Theorem 2(i)) is cited here only as the antecedent of which this is the scalar instance — the proof above does not use it. Certificates: 60,000 random chords with Γ pushed to 10^{-9} of the cone boundary, zero violations (harness; the verifier’s own sweep likewise). Control: with $Q \succ P$ the statement fails — 12,532/20,000 violations, worst +13.2, in the verifier’s sweep; 60,000/60,000 violations, worst –21.4, in the harness’s $Q' = 1.3P$ control. The psd hypothesis is not decoration.

Definition 3 (Admissible frozen filters — a hypothesis with content). A pair (C_0, B_0) is *admissible* if C_0 is monic causal in S and B_0 is causal including lag 0 in R . For such a pair the *minorant*

$$\hat{s}(x; C_0, B_0) = \langle |C_0|^2 f_S + |B_0|^2 \Gamma + 2 \text{Re}(\overline{C_0} B_0 h_1) \rangle$$

is affine in x and satisfies $\hat{s} \geq s$ always.

Remark 22 (R11 — admissibility is a hypothesis, not an adjective). $\hat{s} \geq s$ was checked over 2,880 (record $\times$ frozen-filter) pairs — 120 adversarial records $\times$ 24 filter choices including foreign optima, truncations to 1/2/5/20 lags, oscillating and near-unit-root filters — with $\min(\hat{s} - s) = -2.2 \times 10^{-16}$ (machine zero, attained only where the filter is that record’s own optimum). **A non-monic C ($C = 0.3$) gives $\hat{s} - s = -0.5447$ and inverts the bound**, and fed into the dual it produces “lower bounds” above the true rate. This is the one place where a sign slip inverts the whole certificate.

6.5 The certificate

Theorem 11 (The box-free per-frequency certificate). Let $\mu \geq 0$, let (C_0, B_0) be admissible (Definition 3) with $\hat{s}_0 > 0$ any linearisation point, and set

$$\beta(\omega) = \mu - \alpha |B_0|^2 / \hat{s}_0, \quad p_1 = -\alpha C_0 \overline{B_0} / \hat{s}_0, \quad p_2 = -\mu z^{\Delta+1}.$$

If $\beta > 0$ everywhere then

$$\Psi(D; \Delta) \geq \text{const}(\mu, C_0, B_0, \hat{s}_0) + \left\langle \inf_{h, n \geq 0} \left[\alpha \ln \left(1 + \frac{h R h^*}{n} \right) + \beta (n + h P h^*) + 2 \text{Re}(p^* h) \right] \right\rangle.$$

The three inputs are: (i) admissibility of the frozen filters, (ii) $\beta > 0$, and (iii) a correct cell infimum. In particular the bound requires *no optimality of the fixed point, no convexity of the process-rate functional, no differentiability of the rate, and no moment box*; the separability across frequencies replaces the tangent plane entirely, and Lemma 8 is then used only to *compute* each cell infimum reliably, never to establish validity. *Proof.* Feasibility gives $\text{rate} \geq \text{rate} + \mu(\text{dist} - D)$; Definition 3 gives $-\ln \hat{s} \leq -\ln s$; concavity of $\ln$ gives $-\ln u \geq -u/u_0 - \ln u_0 + 1$; and the result is separable across ω , so the infimum passes inside the average. $\square$

Remark 23 (R17 — state the certificate this way, and why). The dual bound survives brutal mistreatment: 25 random *non-optimal* anchors per Δ give valid bounds; mistuning μ by $\times 0.95$ or $\times 1.10$ degrades but never invalidates; truncating the frozen filters to 12/4/1 lags loses $0/3.6 \times 10^{-6}/1.08 \times 10^{-2}$. The KKT solve only makes the bound *tight*, never *valid*. **This is a genuine strengthening over the moment-box route**, which does *not* survive: a 10^{-4} perturbation of the anchor collapses the box bound to 0.1377. Both agree at the anchor to 2.2×10^{-12} , and **the moment-box computation is therefore recorded as a corroboration, never as the certificate**. Its constants, for the record: $\langle \Gamma \rangle \leq (1 + \sqrt{D})^2 = 2.395445115$ (Minkowski) and $\langle |\Phi_{RV}| \rangle, \langle |\Phi_{RY}| \rangle \leq 1 + \sqrt{D} = 1.547722558$ (pointwise + integral Cauchy–Schwarz, using $\langle f_V \rangle = \langle f_Y \rangle = 1$) — far tighter than the finite- n box, which carries $\sqrt{nD}$ where this carries $\sqrt{D}$ per symbol. A free structural check found en route: $\langle \Gamma \rangle = 1 - D = 0.7$ *exactly* at all three Δ , so the stationary optimum is a proper test channel ($E[\hat{Y}^2] = E[\hat{Y}Y]$). **Consequence: (R2), the global optimality of the stationary program, is discharged — it is no longer required anywhere in the plateau chain.**

Remark 24 (The L^2 box does not exist — constructively). The Hölder pairing above is L^1/L^∞ , and it must be: the distortion constraint controls $\langle \Gamma \rangle$ but *not* $\langle \Gamma^2 \rangle$. Constructively, feasible records at fixed distortion and fixed $\langle \Gamma \rangle$ have $\langle \Gamma^2 \rangle = 1.667/1.965/2.929/6.770$ as the spike narrows over 64/16/4/1 bins — $\langle \Gamma^2 \rangle$ *diverges*. **No L^2 moment box exists here**, and any certificate written against one is invalid.

Remark 25 (Errata and scope notes carried by this section). (R14) — *a proof-step correction*. The cell guard must not be justified by “Hessian $\succeq 2\beta \lambda_{\min}(P)I$ ”: that argument is false, because $\varphi(hRh^*)$ is radially *concave*. The correct strong-convexity modulus is

$$m = \beta \lambda_{\min}(P + Q), \quad \text{guard} = \frac{2|u|^2}{\beta \lambda_{\min}(P + Q)},$$

u the Wirtinger residual at the cell’s stationary point. The corrected modulus is 3.5% larger here (0.14231 against 0.14725) and numerically inert (guards 1.0×10^{-26} against 9.7×10^{-27} ; the bound moves by 0.0×10^0), but the sentence is replaced, not patched.

(R15) — *outward rounding*. The quotable brackets of §6.6 are rounded *outward* at 10^{-10} . The 12-digit line that circulated earlier was rounded *inward* (the lower ends up by $\approx 4 \times 10^{-13}$, the $\Delta = 1$ upper end down by 4×10^{-16}) and is *withdrawn*; only the 10-digit form is quotable. The raw floating-point widths are $\leq 4.4 \times 10^{-12}$.

(R18) — $\Delta = 2$ has no headroom in the tuning. At $\Delta = 2$, $\mu \times 0.95$ costs -2.4×10^{-5} of margin, $\mu \times 1.10$ costs -2.2×10^{-3} , and a $\pm 5\%$ error in the $\hat{s}$ linearisation costs $-4.9 \times 10^{-5} / -1.7 \times 10^{-4}$. Validity is never lost, only margin. $\Delta = 2$ **carries the campaign's smallest margin** and must not be promoted ahead of the other two.

(R19) — *the $O(1/n)$ constant is Δ -dependent*. $n(\phi_{16} - \Psi) = 0.064507/0.070483/0.077503$ at $\Delta = 0/1/2$ (harness: $0.064506/0.070483/0.077503$). Consequently $c(0) \in [0.06447, 0.06449]$ — itself a correction of the earlier “ $0.064495 \pm 3 \times 10^{-5}$ relative”, which was merely the $n = 32$ point; the sequence is monotone decreasing over $n = 8\text{--}48$ with spread 3.0×10^{-4} relative and Richardson(32, 48) = 0.064471 — **is a $\Delta = 0$ statement only. No Δ -uniform $O(1/n)$ constant is claimed anywhere in this document.**

(R20) — *the block infimum*. The anchor-free two-sided certificate of §6.7 pins $\text{block}_\infty \in [0.5299499808, 0.5299499809]$; the value to use is $\text{block}_\infty = 0.5299499808119$. The committed `results/G014-process-limit.json` entry `s6_block_inf` = 0.529949985183839 is **high by 4.4×10^{-9}** and is corrected here. Nothing downstream moves: the 0.0313 spectral gain of Theorem 1(v) is unaffected.

Truncation — and its sign. Lag truncation of the frozen filters is on the *safe* side for the lower bound: a truncated filter is still admissible, so $\hat{s} \geq s$ and the bound can only loosen. It has the *wrong sign* for the upper bound — the rate evaluator's own s is P -truncated, so the computed rate *under-states* the true rate. The upper ends here are safe because the evaluation is flat to 4.4×10^{-16} at the anchors over $P = 50\text{--}1600$; the sign is stated rather than assumed.

6.6 The certified brackets and the plateau

Minimum lower end and maximum upper end over the grid family $N_f \in \{1024, 2048, 4096, 8192\} \times P \in \{60, 100, 140, 200\}$, rounded *outward* at 10^{-10} (R15):

$$\Psi(0) \in [0.5627264963, 0.5627264964], \quad \Psi(1) \in [0.5364013784, 0.5364013785], \quad \Psi(2) \in [0.5310500198, 0.5310500199]$$

The multipliers are $\theta = 3.086038362097/3.130079068857/3.151357871769$, identical to 12 digits across the whole grid family; the lower-end spread over the family is $4.0 \times 10^{-13}/1.4 \times 10^{-12}/7.7 \times 10^{-13}$ and the upper-end spread $\leq 6.7 \times 10^{-16}$. The integrands are analytic and periodic, so the rectangle rule is spectrally accurate (confirmed, not assumed). These are *floating-point* certificates in the sense of Remark 7: no interval arithmetic.

Corollary 7 (The plateau at $\Delta = 0, 1, 2$ — *this supersedes Corollary 5*). *Under Hypothesis 1, Hypothesis 4 and Lemma 6 — and conditional on Lemma 6 alone among the hypotheses this campaign has been fighting —*

$$L^\infty(\Delta) \geq \Psi(D; \Delta) \geq \text{LB}(\Delta),$$

and against the sealed causal-spectral allocation values of `results/G014-process-limit.json`

(s6_cand),

Δ	sealed bar	quotable LB	margin
0	0.5479447799144537	0.5627264963	+ 0.0147817164
1	0.5323438832146611	0.5364013784	+ 0.0040574952
2	0.5302532008457406	0.5310500198	+ 0.0007968190

Every margin clears the certified bracket width by four to eight orders of magnitude ($\Delta = 2$, the tightest, by 1.8×10^8 bracket widths). **This retires (H^*), (H^{**}), (H^{***}), (D), (F) and the boundary charge from the plateau argument, at three lags — two of which the transfer route provably could not reach (it would have needed certified anchors at $n \approx 260$ and $n \approx 1320$).**

Remark 26 (The bars, and a superseded pair of numbers). (R16.) The margins above are computed against the sealed causal-spectral values 0.5479447799144537/0.5323438832146611/0.5302532008457406 of results/G014-process-limit.json (s6_cand), and against nothing else. The Ψ -bracket prover used a rounded bar triple that is correct at $\Delta = 0$ but *wrong* at the other two lags — at $\Delta = 1$ in a *non-conservative* direction, its margin coming out 8.8×10^{-7} too large, and at $\Delta = 2$ conservatively by 8.2×10^{-7} . **That triple is withdrawn: no margin anywhere in this document is computed against it, and none may be.** No verdict moves — all three margins above still clear by four to eight orders — but the $\Delta = 1$ margin recorded against that triple is withdrawn and replaced by +0.0040574952. The harness reads the sealed bars from the committed JSON at run time and cross-checks them against its pinned literals.

Cross-checks, all consistent. An independent window-Cholesky evaluator re-evaluates the certified point to $\leq 1.6 \times 10^{-15}$ at all three Δ and at three window sizes. Every sealed certified LB(ϕ_n) sits above Ψ^{UB} at $n = 8\text{--}48$, with $n(\phi_n^{\text{UB}} - \Psi^{\text{UB}}) = 0.064535/0.064516/0.064507/0.064497/0.064495/0.064487$ — monotone decreasing, exactly as (R19) requires. All independent constructive upper bounds sit above Ψ^{UB} ; a D -ladder over $D = 0.26\text{--}0.34$ sits strictly below the sealed 081 $\phi_8(D)$ lower ends, smooth and monotone; and 400 random feasible stationary records per Δ give zero violations (minimum slack +0.090/ +0.110/ +0.116).

6.7 “Does it prove too much?”

The identical machinery run on the *block* program ($se \equiv n$; no leak, so the cell dual needs *no anchor at all*) returns $\text{UB} = \text{LB} = 0.5299499808119$ against the independently known block infimum — agreement 1.2×10^{-11} , and it does *not* produce a value above the known answer. The full Δ -ladder $0, \dots, 9$ matches the recorded ladder at every Δ ,

$$\Psi(\Delta) = 0.562726496/0.536401378/0.531050020/0.530117530/0.529973408/0.529953058/0.529950369/0.529950029$$

with $\text{LB}(9) - \text{block}_\infty = +7.3 \times 10^{-10} \geq 0$: the machinery approaches the known value *from above* and never crosses it. The per-lag excess ratios rise monotonically *from below* toward $\lambda_s^{-2} = 7.954917$ and cross at $\Delta \approx 6$; the full-family closure constant is $c_{\text{fs}} \approx 0.098 \pm 0.004$, strictly below the diagonal-class $c_{\text{diag}} = 0.111 \pm 0.006$ as it must be. **Keep the two appearances of λ_s apart:** it is the Δ -closure exponent, and it is *not* the kernel pole — the true dominant singularity is a complex pair of modulus $\rho^* = 0.27 \pm 0.02$, uniform in Δ (measured at three Δ), strictly inside $\lambda_s = 0.3545538$, which therefore survives only as a *conservative envelope* for hypothesis (D) (making $\bar{X} = 0.065038$ loose).

6.8 The $\Delta = 2$ finite-window cross-check, certified

The Ψ prover’s scope note — that no *committed* sealed ϕ_n existed at $\Delta = 2$ — is discharged. The same certify machinery that produced the sealed anchors (cold start, deterministic) gives

(n, Δ)	certified bracket for $\phi_n(2)$	width
(16, 2)	[0.535891743313, 0.535893963469]	2.22×10^{-6}
(24, 2)	[0.534274270522, 0.534279127269]	4.86×10^{-6}
(32, 2)	[0.533465073456, 0.533471490973]	6.42×10^{-6}

— at or better than the sealed-cell width band ($3.0\text{--}9.2 \times 10^{-6}$); nothing had to be disclosed as short. Diagnostics: gradient norms $1.19/1.28/1.11 \times 10^{-8}$; the noise-floor parameterization is *inactive* ($\lambda_{\min}(N) \approx 0.1404$ at all three n , and $\lambda_{\max}(N)$ under Theorem 3’s $1/\theta$ cap, so the optimizer is interior); dist $-D$ strictly negative, so each upper end is a *raw feasible* evaluation with zero projection correction; an independent evaluator agrees to 5.6×10^{-16} ; and a second, unrelated cold start lands within 5.3×10^{-7} , inside the certified widths, at all three n . Controls (16, 0), (16, 1), (24, 0) reproduce the sealed brackets first, every upper end within 1.2×10^{-7} .

Cross-check against Corollary 7: no violations. Every certified $\text{LB}(\phi_n(2))$ sits strictly *above* $\Psi(2)$ ’s upper end, by $4.84/3.22/2.42 \times 10^{-3}$ — 500–2000 $\times$ the bracket widths — and $n(\phi_n^{\text{UB}} - \Psi^{\text{UB}}) = 0.0775031/0.0774986/0.0774871$, monotone decreasing as (R19) requires. The committed `s4_fs16_D2` = 0.5358939351814885 lies strictly inside the new (16, 2) bracket.

Remark 27 (Bookkeeping: two valid (16, 0) bracket conventions — this document uses one of them). Two different (16, 0) brackets are in circulation and *both are valid records of different things*: the R-IND-5 *width-of-record* bracket $[v^* - 3.45 \times 10^{-6}, v^*] = [0.566754682, 0.5667581350]$, and the `results/G014-convexity.json` bracket produced by the harness’s own cold start. **This document uses the *width-of-record* convention throughout** — it is the convention already printed in the anchor table of §3.2 — and every (16, 0) bracket quoted anywhere above is that one. Mixing the two in a single comparison is an error.

Hygiene notes carried forward. All brackets in this section are floating-point certificates (Remark 7); the finite- n lower endpoints are BLAS-sensitive at $\sim 10^{-7}$ through $r_n \cdot R_{\text{box}}$, which is exactly why the width gates live in the artifact-self-consistency tier of CI and why a governed runner must *re-derive* these numbers and never copy them. Two artifact-hygiene items are on record and are not cited as evidence anywhere above: the committed `spectral/transfer.log` kernel-transfer defect is an A_v lag-orientation mismatch (a flat $6.76 \times 10^{-2} = |a_v(+1) - a_v(-1)|$ swamping a genuine $O(1/n)$ boundary layer) and must be regenerated before use; and the internal check “block + leak = rate” is an *algebraic identity*, not an independent validation.

Question 1 (Open). (0) *Lemma 6, and it is now the single remaining load-bearing hypothesis of the plateau (with Hypothesis 4, which is a hypothesis of an explicit construction rather than an estimate). It is a pointwise-limit-of-convex-functions statement on the cyclostationary class; a proof would leave Corollary 7 standing on Hypothesis 1 alone. Separately, the reverse inequality in Corollary 6 — a truncation / achievability lemma at stationarity, which the non-rationality of the stationary optimum makes non-trivial — is open, and until it is settled nothing may be said about L^∞ beyond the bracket of Remark 20; (1) A uniform-in- m decay bound on the optimizer’s cross-block kernel — no longer needed for the plateau (Corollary 7 retires it there),*

but still the residue of the old n -monotonicity face after Theorem 4 closed its subadditivity half unconditionally and Theorem 5 reduced the rest to the single scalar X . The open problem is now named, and it is not “bound $I(E; T^{b_2})$ ”: (i) no uniform family bound can exist (Remark 12); (ii) no KKT/exchange argument can force $X = 0$, because the distortion gradient vanishes identically on the cross-read coordinates (Theorem 3) — the cross-read is a distortion-free direction on which the objective is stationary and the optimizer chooses $X > 0$ because it strictly lowers L_a , so stationarity yields an equation, not a bound; (iii) what is missing is an optimizer decay estimate: measured, $\|A^{12}\|_F / \|A^{11}\|_F = 3.44\%$ with the cross-block kernel decaying geometrically at per-lag ratio $\approx 0.28 - 0.35$, bracketing the smoothing pole $\lambda_s = 0.354554$, and $\|A^{12}\|_F$ is flat in m . A Demko/Jaffard-type exponential off-diagonal decay bound for inverses of banded positive-definite operators, applied to the KKT system of Theorem 3, would give $\|A^{12}\| \leq C\lambda^\bullet$ uniformly in m and close the gap — a standard type of theorem, much softer than (H^{**}) . The conversion from such a bound to $\bar{X}$ is already proved (Remark 17); the two pieces still missing are the fixed-point/bootstrap step that breaks the circularity $M = N^{-1} - \theta I$, and the uniform lower bound $\lambda_{\min}(N) \geq \nu$. Alternatively (H^{***}) (Remark 11) may be attacked directly, since the split-causality value gap is two-sided certifiable. Separately, certified anchors at $n \approx 260/1320$ would carry the $\Delta = 1, 2$ plateaus without any of the three hypotheses; and the n -uniform dual certificate of the GO-13 spectral machinery would bypass the transfer route altogether; (2) per-cell convexity (Remark 4) — the within-class certification of the diagonal ladder (Remark 5) is no longer open, it is executed in §5.2; (3) the U -coupled coordinate (Hypothesis 1 dropped): a different, far cheaper object, definitionally open; (4) the reset protocol realizing definition (a); (5) identification of the closure constant beyond $\Delta \approx 6$, and the innovations characterization of the limit (vector NRDF closed forms were refuted in general — Stavrou–Tanaka–Tatikonda — so only scalar forms are expected cheaply).

House rules: the novelty gate is *cleared* for the four objects swept (arXiv + DBLP + Cross-ref; S2 residual is belt-and-suspenders only), conditional on the five must-cite adjacents (Berta et al.; Boyd–Mandal–Crutchfield; Venkat–Weissman–Carmon–Shamai; Sandberg et al.; Anderson 2018) — headline novelty language may be promoted only with the five citations in force, and nothing above claims it yet. R-IND-5 passes 1 (+closure), the Conjecture-v0.2 pass, the Theorems-R+C pass (seven mandatory restatements) and the n -transfer pass (six mandatory restatements — the (H^*) demotion, the κ -per-side refutation, the constants, the $\mathcal{F}_0$ -only tags, the corollary arithmetic and the ladder tagging — all in force in this revision) are on record (go14-causal-erasure-PROBE.md); seal GO-P-2026-076 governs the theorem’s numerical faces; the process-limit faces of §4 await prereg 078 (harness `experiments/go14_process_limit.py`); the convex-program faces of §3 await prereg 079 (harness `experiments/go14_convexity.py`: representation identity incl. non-staircase schedules, PSD/closed-form/Sylvester certificates, 4,200-pair Jensen sweep, tangent plane, cold-start certified brackets at $(16, 0)/(16, 1)/(24, 0)/(32, 0)$ + block_{16} , sandwich-margin window, and the two-leg / non-conflation U -gates); the transfer faces of §5 await prereg 080 (harness `experiments/go14_transfer.py`: per-cell subadditivity with block-local denominators, the interleaved-prefix set identity, the $\mathcal{F}_0$ -only zero-claim with its U -coupled counter-values, the (H^*) refutation netted as a gated fact, the optimizer charges at $m \in \{8, 16\}$, the constants and the corollary arithmetic including the failing $n = 24$ base, and reproduction of the transfer anchors and of the within-class ladder). The scope proposed for 080 is: Theorem 4 and Corollary 2 (unconditional), the transfer anchors, the within-class ladder, and the conditional Theorem 7 / Corollary 5.

The v0.6 material — Theorem 3 and its corollaries (§3.1), the H-repair lemma chain L1–L4

and Theorem 5, the split-causal sub-family with its certified ϕ^{sc} brackets and Theorem 6, the Π -retraction identities (Lemma 5), the extended monotone $m \times \Delta$ table, the convexity-of- $\phi_m(D)$ hygiene of Remark 14, the recomputed corollary arithmetic under (H^{**}) and under $(D)+(F)$, and the (H^{***}) value gaps — awaits **registration 081**, harness `experiments/go14_hrepair.py` (s1 the L1 identity and its m -independence; s2 the L2 bound with its tightness witness; s3 L3/L4 and the H-repair inequality over a 22-cell pinned battery including both refuting counterexamples; s4 the split-causal sub-family, its constants and the linearity of the section; s5 the Π -retraction identities; sK the three Theorem-K identities, the two structural facts, the Cholesky split, A_y 's transpose symmetry and the m -uniform bounds; s6 the block-distortion straddle and the value-function convexity; s7 the constants and both corollary arithmetics with the base-24 negative control; s8 the monotone m -table; s9 $\phi_{2m}^{\text{sc}} \leq \phi_m$ and the (H^{***}) gaps). Nothing in §3.1 or §5.1 is sealed.

The v0.7 material — §6 in full: Theorem 8 with Remark 18, Theorem 9 with Hypothesis 4 and Remark 19, Lemma 6, Theorem 10, Corollary 6 with Remark 20, Lemma 7, Lemma 8, Definition 3 with Remark 22, Theorem 11 with Remarks 23 and 24, the errata of Remark 25 (R14, R15, R18, R19, R20 and the truncation sign), the certified brackets of §6.6, Corollary 7 with Remark 26, §6.7, the $\Delta = 2$ anchors of §6.8 and the bookkeeping convention of Remark 27 — awaits **registration 082**, harness `experiments/go14_plateau.py` (s1 the Collapse identity on staircase, edge and non-staircase schedules with the R6 Δ -lag-causal U -extension gated in *both* directions; s2 the three sub-claims of step (A) with the R7 correlated-noise-copy control; s3 step (B) at cold-start optimizers and at random period- n records — cone membership, distortion preservation, and the no-larger-rate inequality; s4 Lemma 8's Jensen sweep at 10^{-9} from the cone boundary with the R12 $Q \succ P$ control; s5 the minorant over 2,880 record $\times$ filter pairs with the R11 non-monic control; s6 R17 — validity of the dual at deliberately non-optimal anchors, mistuned μ , truncated filters and mistuned linearisation; s7 reproduction of the quotable brackets, of the margins against the *sealed* bars, of block_∞ (R20) and of the Δ -dependent $O(1/n)$ constants (R19); s8 the “does it prove too much” block program and the Δ -ladder). The scope proposed for 082 is: Theorem 8, Theorem 9 and Theorem 11 (unconditional given Hypothesis 1 and Hypothesis 4); Lemmas 7 and 8; the certified Ψ brackets; the $\Delta = 2$ finite- n anchors; and Theorem 10 / Corollary 6 / Corollary 7 *conditional on Lemma 6*. Nothing in §6 is sealed, and no seal may print an equality between L^∞ and Ψ (Remark 20).